A DICHOTOMY FOR ERDŐS PROBLEM 1112: EXISTENCE OF $r_k(d_1, d_2)$ FOR $k \geq 3$

JOHAN LAND

Abstract. Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers. Erdős problem 1112 asks whether there is an integer r such that every lacunary sequence $B = \{b_1 < b_2 < \cdots\}$ of positive integers with $b_{i+1} \geq r\, b_i$ admits a sequence $A = \{a_1 < a_2 < \cdots\}$ of positive integers with all gaps in $[d_1, d_2]$ and $(kA) \cap B = \emptyset$, where kA is the k-fold sumset; write $r_k(d_1, d_2)$ for the least such r.

We prove that $r_k(d_1, d_2)$ exists if and only if $d_2 \geq k+1$ — so the answer to the recorded question is yes precisely when $d_2 \geq k+1$ — with the explicit bound $r_k(d_1, d_2) \leq 192\, d_2$ when it exists. Below the threshold, non-existence holds in a strong form: for every prescribed sequence of pointwise lower bounds on the successive ratios of B, however rapidly growing, a single B meets the k-fold sumset of every admissible A.

The existence half uses a Beatty construction whose k-fold sumset threads the gaps a lacunary B must leave; the non-existence half reduces, through a word-combinatorial analysis of the gap sequence, to a bounded subset-sum covering lemma of possible independent interest. The proof is computer-assisted, with a finite certificate layer, and the full dichotomy is formally verified in Lean 4 in the problem's literal integer phrasing.

1. Introduction

1.1. The problem.

1.1. The problem. Erdős problem 1112 [E1112] asks, verbatim:

> Let $1 \leq d_1 < d_2$ and $k \geq 3$. Does there exist an integer r such that if $B = \{b_1 < \cdots\}$ is a lacunary sequence of positive integers with $b_{i+1} \geq rb_i$ then there exists a sequence of positive integers $A = \{a_1 < \cdots\}$ such that $d_1 \leq a_{i+1} - a_i \leq d_2$ for all $i \geq 1$ and $(kA) \cap B = \emptyset$, where kA is the k-fold sumset?

We pause over the phrase "does there exist an integer r". For a *negative* r the condition $b_{i+1} \geq rb_i$ is vacuous, since B is positive; and $r \leq 1$ never works, because then $B = \{1, 2, 3, \ldots\}$ is admissible and meets every nonempty kA. So the existence question over $\mathbb{Z}$ is equivalent to the same question over the integers $r \geq 2$, and any witness is automatically at least 2. Accordingly we define

$$r_k(d_1, d_2) \;:=\; \text{the least integer } r \geq 2 \text{ with the stated property,}$$

which is well defined precisely when the property holds for some r; "$r_k(d_1, d_2)$ exists" abbreviates that. The quantity $r_k(d_1, d_2)$ was introduced by Bollobás, Hegyvári and Jin [BHJ97]; we use the problem page's integer-r convention throughout. (The formalization quantifies r over $\mathbb{N}$; see §6.)

Throughout, A and B are infinite sets of positive integers, identified with their strictly increasing enumerations, and

$$kA \;=\; \{x_1 + \cdots + x_k \;:\; x_1, \ldots, x_k \in A\}$$

Date: July 2026.

2020 *Mathematics Subject Classification.* 11B13, 11B75, 11P70, 68V20.

Key words and phrases. Erdős problems, lacunary sequences, sumsets, Sturmian words, subset sums, formal verification, Lean.

Source, data and the Lean development: `https://github.com/beetree/math_erdos_1112`. See §9 for the release and reproduction instructions, and §7 for a statement of how this work was produced.

is the k-fold sumset *with repetitions allowed*. We call a sequence A with $d_1 \le a_{i+1} - a_i \le d_2$ for all i a (d_1, d_2)*-sequence*. Explicitly, "$r_k(d_1, d_2)$ exists" means: there is an integer r such that *every* lacunary B of ratio r admits *some* (d_1, d_2)-sequence A with $(kA) \cap B = \emptyset$; the order of quantifiers matters — A is chosen after B.

Remark 1.1 (Attribution). We solve the formulation recorded as Erdős problem 1112 on `erdosproblems.com`. As that page itself notes, the formulation is a generous interpretation of an ambiguous remark of Erdős and Graham [ErGr80, p. 18]; it may be more accurate to describe it as a problem *inspired by* them. Our results concern the formulation as stated there, which is the one quoted above.

1.2. **Prior results.** The problem page records the following. Erdős–Graham [ErGr80] and Bollobás–Hegyvári–Jin [BHJ97] give $r_2(2, 3) = 2$; Chen [Ch00] shows $r_2(a, b) \le 2$ for integers $a < b$ with $b \ne 2a$ (and $r_2(a, 2a) \ge 2$, nontrivial only in the real-ratio convention of that paper); and [BHJ97] shows that $r_3(2, 3)$ does not exist, in a strong form. Tang and Yang [TaYa21] prove, for $k \ge 3$, that if the difference sequence of A is restricted to a structured class (*block type*) then there is a B with $(kA) \cap B \ne \emptyset$ — in the spirit of our eventually periodic case (Lemma 3.6); the difficulty in the general problem lies precisely in the non-periodic (Sturmian) tails, which we handle in §3. The general existence question for $k \ge 3$ is listed as open; as of July 2026 the problem page records no further progress on it, and we know of none elsewhere.

1.3. **Results.** Our main result answers the question completely.

Theorem A (The dichotomy). Let $k \ge 3$ and $1 \le d_1 < d_2$. Then:

(1) (Existence.) If $d_2 \ge k + 1$ then $r_k(d_1, d_2)$ exists; explicitly, $r_k(d_1, d_2) \le 192\, d_2$.
(2) (Non-existence, strong form.) If $d_2 \le k$ then $r_k(d_1, d_2)$ does not exist. Indeed, for *every* prescribed sequence $(r_i)_{i \ge 1}$ of nonnegative integers — with no monotonicity assumed — there is a *single* B with $b_{i+1} \ge r_i b_i$ for all i such that $(kA) \cap B \ne \emptyset$ for *every* (d_1, d_2)-sequence A. (Real lower bounds reduce to this: replace each r_i by $\lceil r_i \rceil$.)

Hence $r_k(d_1, d_2)$ exists if and only if $d_2 \ge k + 1$, and the answer to the quoted question is: *yes precisely when* $d_2 \ge k + 1$.

Why does the threshold sit at $d_2 = k + 1$? Summing k terms of a sequence with average gap $\gamma \le d_2$ confines kA, up to accumulated rounding, to windows of width k hanging below the multiples of γ (this is (3) below). If $d_2 \ge k + 1$, we may *choose* A with $\gamma \in (k, d_2)$: consecutive windows then leave gaps of width $\gamma - k > 0$, a fixed positive proportion of every scale, and the construction's only job is to steer γ so that a lacunary B falls into the gaps. If $d_2 \le k$, no admissible A can spread out this way: every gap value is at most k, and k summands turn out to be enough freedom to fill an entire congruence class — at the binding corner $k = d_2$, exactly enough, which is where the subset-sum lemma below is spent. The two halves make these two sentences precise.

Note that d_1 is irrelevant to the threshold, and this is visible in both halves of the proof. The existence construction uses only the two largest admissible gaps, $d_2 - 1$ and d_2, which lie in $[d_1, d_2]$ whatever d_1 is. The non-existence argument works with the gap values that recur infinitely often; their *maximum* is what the budget of k summands must pay for, again independently of d_1 (which enters only through $d_1 \ge 1$). The constant 192 is not optimized, and we make no claim about the true order of $r_k(d_1, d_2)$. Part (2) is genuinely stronger than the negation of the question: no prescribed sequence of *pointwise lower bounds on the successive ratios* of B — however rapidly growing, and fixed in advance — can rescue the existence of A. The quantification is over ratio sequences (r_i), not over other structural or arithmetic conditions one might impose on B.

Corollary 1.2 (Consecutive gap windows)**.** *Let $k \geq 3$ and $d \geq 1$. Then $r_k(d,\, d+1)$ exists if and only if $d \geq k$.*

This is the $d_2 = d+1$ slice of Theorem A.

Remark 1.3 (The case $k = 2$, and why §3 needs $k \geq 3$). Theorem A is stated for $k \geq 3$, as is the problem. For context, $r_2(2,3) = 2$ [ErGr80, BHJ97], and $r_2(a,b) \leq 2$ for $a < b$ with $b \neq 2a$ [Ch00] (in the convention of [Ch00], which permits real ratios; under the integer convention of this paper a witness is automatically ≥ 2). The strong form of non-existence is *not* claimed at $k = 2$, and the hypothesis $k \geq 3$ in §3 is not an artifact of the method: the width machinery does not extend to $k = 2$. Its engine, Lemma 3.12, converts a single two-index antidiagonal of width ≥ 2 into a width bound for all large s; at $k = 2$ there are balanced words whose two-index width W_2 vanishes infinitely often, leaving that lemma nothing to work with. (We do not pursue $k = 2$, and make no claim there.) The existence half (Theorem 2.1), by contrast, is proved here for all $k \geq 2$.

The non-existence half turns on a self-contained combinatorial fact about subset sums, which we isolate as a lemma and prove in §4. For a finite multiset S of positive integers write $\text{subset-sums}(S) := \{\sum_{x \in T} x : T \subseteq S\}$ (a selection of a sub-multiset), and for a finite set G of positive integers with $M := \max G$ put

$$m(G) \;:=\; \min\bigl\{\, |S| \;:\; S \text{ a multiset drawn from } G,\ \text{subset-sums}(S) \supseteq \text{an interval of } M \text{ consecutive integers} \,\bigr\}. \tag{1}$$

Lemma 1.4 (Bounded subset-sum covering)**.** *For every finite $G \subset \mathbb{Z}_{\geq 1}$ with $|G| \geq 3$, $\gcd(G) = 1$ and $\max(G) = M$, we have $m(G) \leq M - 1$.*

We should say precisely what this bound does and does not assert. It is *not* the claim that $M-1$ is the exact value of $m(G)$ for every G; typically $m(G)$ is far smaller. Rather, $M-1$ is (i) exactly the budget that the non-existence argument consumes at its binding corner $k = d_2 = M$ (see the discussion after Lemma 3.18), and (ii) not improvable as a universal bound:

Proposition 1.5 (Non-improvability)**.** *Both hypotheses are used to their limit. The constant $M-1$ cannot be replaced by $M-2$ (indeed $m(\{3,4,5\}) = 4 = M-1$), and the cardinality hypothesis $|G| \geq 3$ cannot be dropped: for $G = \{2,3\}$ (so $\gcd = 1$, $M = 3$) no multiset of size $\leq M - 1 = 2$ has 3 consecutive integers among its subset sums.*

The proof is a small finite check, deferred to §4.10 to keep the introduction moving.

1.4. **Relation to the literature.** Although we use Lemma 1.4 only as a tool, it may be of independent interest, so we place it briefly against three traditions.

Dense-set subset-sum theorems. The results that produce long intervals or progressions among subset sums — Sárközy [Sa89, Sa94], Lev [Lev03], Szemerédi–Vu [SV06], Conlon–Fox–Pham [CFP21], Alon–Freiman [AF88], and constructively Chen–Mao–Zhang [CMZ25] — work with a *dense* set ($|A| \gtrsim \sqrt{n}$) and yield an arithmetic progression whose common difference need not be 1.

The closest antecedent is Lev's consecutive-subset-sum theorem [Lev98, Thm. 1], which does give a genuine interval of consecutive integers among the subset sums of a set — but again only a *dense* one ($|A| \gtrsim \frac{2}{3}M$), vacuous when $|G| = 3$. Lemma 1.4 is the *density-free, multiset* analogue: it trades density for repetition, covering an interval of length exactly M at uniform cost $|S| \leq M - 1$ from an arbitrary gcd-1 set down to $|G| = 3$.

Higher-sumset structure. The structure theorems for m-fold sumsets (Nathanson [Na72], Granville–Walker [GW21], Lev [Lev22]) are density-free but describe mG with m large — a total-degree "simplex" of coefficients, of which the subset sums of a bounded multiset are a per-generator "box" inside — so their conclusion is a weaker statement about a larger set.

Completeness and Frobenius. The completeness, Frobenius and postage-stamp traditions [Gr64, ErGr80, RA05, GS20] concern infinite sequences, or use unboundedly many summands, or optimise the generating set; none bounds a multiset drawn from an arbitrary given set.

We are not aware of a result that implies Lemma 1.4, even with a worse constant, and after a documented search[1] it appears to be new. As with any negative claim in a mature area this warrants a specialist's confirmation; the sharpest place to look is the neighbourhood of Lev's consecutive-subset-sum theorem [Lev98].

1.5. **Method.** The two halves of Theorem A are proved by unrelated methods.

For *existence* (§2) we place A inside a Beatty family $A_{\gamma,c} = \{c + \lfloor i\gamma \rfloor : i \geq 1\}$ with γ slightly below d_2. Then the gaps of A automatically lie in $\{d_2 - 1, d_2\}$, and accumulated floor error confines kA to a union of half-open windows of width k hanging below the multiples of γ. Avoiding a given b becomes a statement about γ avoiding a union of "unsafe" intervals, and the crux (Lemma 2.2) is that one can *name a single gap* between consecutive unsafe intervals rather than enumerate them. A nested-interval argument then defeats all of B at once, and the lacunarity ratio $192\,d_2$ is exactly what keeps the nested intervals from collapsing.

For *non-existence* (§3) the target is the property that kA is *tail-covering*: it eventually contains a full congruence class. A diagonalization (Lemma 3.1) shows that a single B then defeats every A simultaneously, so everything reduces to proving that every (d_1, d_2)-sequence with $d_2 \leq k$ is tail-covering. This is done by cases on the *tail alphabet* G_∞ of gap values occurring infinitely often. One and two letter alphabets require a word-combinatorial analysis culminating in the Sturmian case (Lemma 3.14); alphabets with three or more letters are handled by an amortized slot argument (Lemma 3.18) that reduces matters exactly to Lemma 1.4, proved in §4 by a six-branch decision tree.

1.6. **Formal verification and the trust base.** The complete dichotomy — all three theorems, including the strong non-existence form — has been formalized in Lean 4 and machine-checked. The development is `sorry`-free and depends only on Lean's three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), with no `native_decide`, so every finite check is discharged by the trusted kernel. The argument separates into three auditable layers: the **prose** proof of §§2–5, which carries all infinite content; the **finite layer** — the certificate tables and bounded scans, isolated as Proposition A.1, each row a self-certifying witness (also re-checked by two independent Python harnesses and re-decided in the kernel); and the **Lean development**, certifying both together. Machine-checking establishes correctness relative to the formal statement; it cannot establish novelty or priority, which rest on §1.4. Section 6 gives the formal encoding and trust base, and Appendix C maps each result to its Lean declaration. The division of labour with the automated systems used is set out in §7.

1.7. **Map of the proof.** The two halves are unrelated, and the reader may take them in either order.

[1]Conducted July 2026 over MathSciNet, zbMATH, arXiv and Google Scholar citation chains around the works cited here; the record, with a full comparison table, is in the repository (`paper/novelty-search.md`).

Theorem A ($r_k(d_1, d_2)$ exists $\iff$ $d_2 \geq k+1$)	
$d_2 \geq k+1$ (existence, §2)	Beatty family $A_{\gamma,c}$ (§2) $\to$ free-gap lemma (Lemma 2.2) $\to$ nested intervals $\to$ ratio $192d_2$ (Theorem 2.1)
$d_2 \leq k$ (non-existence, §3)	certificate lemma (Lemma 3.1): one B defeats all A, *provided* every admissible A is *tail-covering.* Then split on the tail alphabet G_∞: $\lvert G_\infty\rvert = 1$: arithmetic progression (Lemma 3.4) $\lvert G_\infty\rvert = 2$: eventually periodic (Lemma 3.6), or k even, $d_2 = k \to$ boundary routing (Prop. 3.10), or width $\geq 2 \to$ sweep (Lemmas 3.9, 3.8), or balanced $\to$ Prop. 3.11 $\to$ Sturmian (Lemma 3.14) $\lvert G_\infty\rvert \geq 3$: slot lemma (Lemma 3.18), which consumes **Lemma 1.4** (§4, six branches D/P/L/E/T/B)

The only point at which §4 enters the proof of Theorem A is through the slot lemma of the non-existence half: Lemma 1.4 is what that lemma needs at its binding corner $k = d_2 = M$, and it is where the finite certificate data of Appendix B enters.

Three reading routes follow from this structure. A reader who wants the dichotomy may read §§2, 3 and 5, taking Lemma 1.4 on faith. §4 is self-contained additive combinatorics and can be read independently, starting from its own roadmap. The verification material — §6, the appendices, and the supplement — is reference matter, needed for audit rather than for the mathematics.

1.8. **Notation.** For a (d_1, d_2)-sequence $A = \{a_1 < a_2 < \cdots\}$ we write $g_n := a_{n+1} - a_n \in [d_1, d_2] \cap \mathbb{Z}$ for its *gaps*, and $p_0 := 0$, $p_n := g_1 + \cdots + g_n$ for the *prefix sums*, so that $a_{n+1} = a_1 + p_n$ and

$$kA \;=\; ka_1 + \{\, p_{i_1} + \cdots + p_{i_k} \;:\; i_1, \ldots, i_k \geq 0 \,\}. \tag{2}$$

The *gap word* of A is the sequence $(g_n)_{n\geq 1}$, and G_∞ — the *tail alphabet* — denotes the set of gap values occurring infinitely often in it. We call A *tail-covering* if

$$kA \;\supseteq\; \{x \geq X_0 \;:\; x \equiv \rho \pmod{m}\} \qquad \text{for some } m \geq 1,\ \rho,\ X_0.$$

Intervals of integers are written $[u, v] = \{u, u+1, \ldots, v\}$; for a real interval we say so explicitly. The quantity $m(G)$ is as in (1).

2. Existence for $d_2 \geq k+1$

Theorem 2.1. *Let $k \geq 2$, $d_2 \geq k+1$ and $1 \leq d_1 < d_2$. Then $r_k(d_1, d_2) \leq 192\, d_2$.*

The proof occupies the rest of this section. Fix B with $b_{i+1} \geq rb_i$, where $r := 192\, d_2$. We construct A inside a Beatty family.

2.1. **Beatty cluster containment.** For real $\gamma \in (d_2 - 1, d_2)$ and an integer $c \geq 0$ set

$$A_{\gamma,c} \;:=\; \{\, c + \lfloor i\gamma \rfloor \;:\; i \geq 1 \,\}.$$

Its consecutive gaps are $\lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor \in \{\lfloor \gamma \rfloor,\ \lfloor \gamma \rfloor + 1\} = \{d_2 - 1, d_2\} \subseteq [d_1, d_2]$ (note $\lfloor \gamma \rfloor = d_2 - 1$, as $d_2 - 1 < \gamma < d_2$), so $A_{\gamma,c}$ is a (d_1, d_2)-sequence for every such γ and c. For any $i_1, \ldots, i_k \geq 1$ with $s := \sum_t i_t$ we have, from $i_t\gamma - 1 < \lfloor i_t\gamma \rfloor \leq i_t\gamma$,

$$\sum_t \lfloor i_t\gamma \rfloor \;\in\; (\, s\gamma - k,\; s\gamma \,],$$

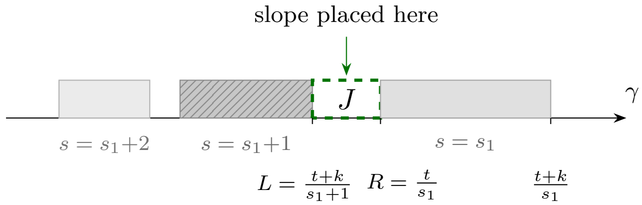

FIGURE 1. (Lemma 2.2.) For a fixed shifted target $t = b - kc$, the unsafe intervals $[t/s,\ (t{+}k)/s]$ on the slope axis: higher denominators s give intervals further left. The free gap is $J = (L, R)$, where $L = (t{+}k)/(s_1{+}1)$ is the upper endpoint of the $s_1{+}1$ interval and $R = t/s_1$ the lower endpoint of the s_1 interval; every interval with $s > s_1$ lies left of J and every one with $s \le s_1$ lies right of it, and the construction places its slope in the closed middle third of J.

and therefore

$$kA_{\gamma,c} \ \subseteq\ \bigcup_{s\ge k} \big(\, kc + s\gamma - k,\ kc + s\gamma \,\big]. \tag{3}$$

Consequently, if for a given b no integer $s \ge 1$ satisfies $s\gamma \in [\, b - kc,\ b - kc + k \,]$, then $b \notin kA_{\gamma,c}$. Two deliberate weakenings make this criterion easier to verify than (3) requires. First, we quantify over all $s \ge 1$ rather than only the $s \ge k$ that actually occur in (3); excluding more values of s is a stronger — hence safe — demand, and it frees the argument from tracking which s are realizable. Second, we use the *closed* interval $[\, b - kc,\ b - kc + k \,]$ in place of the half-open window of (3), which again only strengthens the requirement. No irrationality of γ is needed anywhere.

2.2. **The free-gap lemma.** This is the engine of the construction. The idea (Figure 1) is geometric: the slopes γ that would let $kA_{\gamma,c}$ hit a given element of B form a sparse union of *unsafe* intervals $[t/s, (t+k)/s]$, and instead of counting them we simply exhibit one explicit gap between consecutive unsafe intervals — wide enough to hold a usable slope.

Lemma 2.2 (Free gap). *Let $t \ge 32d_2^2$ and let $I \subseteq [\, d_2 - \frac{1}{2},\ d_2 \,]$ be a closed real interval with $|I| \ge 4d_2^2/t$. Then I contains a closed subinterval I' with $|I'| \ge d_2/(48t)$ such that every $\gamma \in I'$ satisfies $s\gamma \notin [t, t+k]$ for all integers $s \ge 1$.*

Proof. Write $I = [lo, hi]$, so $d_2 - \frac{1}{2} \le lo \le hi \le d_2$ and $hi - lo \ge 4d_2^2/t$. A real γ satisfies $s\gamma \notin [t, t+k]$ for all $s \ge 1$ exactly when γ avoids the *unsafe set*

$$U \ :=\ \bigcup_{s\ge 1} [\, t/s,\ (t+k)/s \,].$$

Rather than count the unsafe intervals meeting I, we name a single gap between two of them. Put

$$s_1 := \lceil t/hi \rceil \quad \text{(the least } s \text{ with } t/s \le hi\text{)}, \qquad R := t/s_1, \qquad L := (t+k)/(s_1+1),$$

and let $J := (L, R)$, the gap between the s_1-th and $(s_1{+}1)$-st unsafe intervals. From $t/hi \le s_1 < t/hi + 1$, together with $hi \in [d_2 - \frac{1}{2}, d_2]$ and $t \ge 32d_2^2$, one gets

$$\frac{t}{d_2} \le s_1, \qquad s_1 \ge 32d_2, \qquad s_1 + 1 \le \frac{5t}{d_2}. \tag{4}$$

(i) J avoids U. If $s \le s_1$ then $t/s \ge t/s_1 = R$, so any $\gamma < R$ has $s\gamma < t$. If $s \ge s_1 + 1$ then $(t+k)/s \le (t+k)/(s_1+1) = L$, so any $\gamma > L$ has $s\gamma > t + k$. Hence a single $\gamma \in J$ misses *every* interval $[t/s, (t+k)/s]$ at once; no enumeration of a relevant range of s is required.

(ii) $|J| \geq d_2/(16t)$. We have $R - L = (t - ks_1)/(s_1(s_1+1))$. Since $s_1 \leq t/hi + 1 \leq t/(d_2 - \frac{1}{2}) + 1$ and $k \leq d_2 - 1$,

$$\Big(k + \tfrac{15}{32}\Big)s_1 \leq \Big(d_2 - \tfrac{17}{32}\Big)\Big(\frac{t}{d_2 - \frac{1}{2}} + 1\Big)$$
$$= t - \frac{t}{32(d_2 - \frac{1}{2})} + \Big(d_2 - \tfrac{17}{32}\Big) \;\leq\; t,$$

the last step because $t \geq 32d_2^2$ gives $t/\big(32(d_2 - \frac{1}{2})\big) = d_2^2/(d_2 - \frac{1}{2}) \geq d_2 + \frac{1}{2} > d_2 - \frac{17}{32}$. Thus $t - ks_1 \geq \frac{15}{32}s_1$, and with (4),

$$R - L \;\geq\; \frac{15}{32} \cdot \frac{1}{s_1 + 1} \;\geq\; \frac{15}{32} \cdot \frac{d_2}{5t} \;=\; \frac{3}{32} \cdot \frac{d_2}{t} \;\geq\; \frac{d_2}{16t}.$$

(iii) $J \subseteq I$. First $R = t/s_1 \leq hi$ by the choice of s_1, and $s_1 < t/hi + 1$ gives $R > t/(t/hi + 1) > hi - hi^2/t \geq hi - d_2^2/t$. Also $R - L \leq t/(s_1(s_1+1)) \leq t/(t/hi)^2 \leq d_2^2/t$, so

$$L = R - (R - L) \;\geq\; (hi - d_2^2/t) - d_2^2/t \;=\; hi - 2d_2^2/t \;\geq\; hi - \tfrac{1}{2}(hi - lo) \;\geq\; lo.$$

Hence $[L, R] \subseteq [lo, hi] = I$.

Now take I' to be the closed middle third of J. Then $I' \subseteq I$, every $\gamma \in I'$ avoids U by (i), and $|I'| = |J|/3 \geq d_2/(48t)$ by (ii). □

2.3. **Small elements of B, and the nested intervals.** Since B is infinite and increasing, we may fix an index j_0 with $b_{j_0} \geq 4d_2^2$. Set $c := b_{j_0} + 1$. Then $\min kA_{\gamma,c} \geq k(c + d_2 - 1) > kc > b_{j_0}$, so every $b_i \leq b_{j_0}$ lies below $\min kA$ and is avoided automatically. For $i > j_0$ put $t_i := b_i - kc$; these are positive, since $b_i \geq b_{j_0+1} \geq 192d_2b_{j_0}$ while $kc = k(b_{j_0}+1) \leq d_2(b_{j_0}+1)$, and $192d_2b_{j_0} > d_2(b_{j_0}+1)$ for $b_{j_0} \geq 1$. Since $(x - \tau)/(y - \tau) \geq x/y$ whenever $x \geq y > \tau \geq 0$, the shifted targets retain the ratio:

$$\frac{t_{i+1}}{t_i} \;\geq\; \frac{b_{i+1}}{b_i} \;\geq\; 192d_2,$$

and moreover $t_{j_0+1} = b_{j_0+1} - kc \geq 192d_2b_{j_0} - d_2(b_{j_0}+1) \geq 96d_2b_{j_0} \geq 384d_2^3 \geq 32d_2^2$.

Proof of Theorem 2.1. Start with the closed real interval $I_0 := [\,d_2 - \frac{1}{2},\; d_2 - \frac{1}{4}\,]$, which has $|I_0| = \frac{1}{4} \geq 4d_2^2/t_{j_0+1}$ by the previous paragraph. Applying Lemma 2.2 with $t = t_{j_0+1}$ produces a closed $I_1 \subseteq I_0$, safe for t_{j_0+1}, with $|I_1| \geq d_2/(48\,t_{j_0+1})$. Now induct for $j \geq 1$: suppose I_j is safe for the targets $t_{j_0+1}, \ldots, t_{j_0+j}$ and satisfies $|I_j| \geq d_2/(48\,t_{j_0+j})$ (both well defined, as $j \geq 1$). Apply Lemma 2.2 with $t = t_{j_0+j+1}$: its hypothesis $|I_j| \geq 4d_2^2/t$ reduces to

$$t_{j_0+j+1} \;\geq\; 192\,d_2\,t_{j_0+j},$$

which is exactly what the ratio $r = 192d_2$ supplies. This is the constant's entire ledger: $192 = 4 \times 48$, the hypothesis constant of Lemma 2.2 (an interval of width $4d_2^2/t$ is wide enough) against its output width ($d_2/(48t)$ survives), so each application leaves exactly enough room for the next. This produces a closed $I_{j+1} \subseteq I_j$, safe also for t_{j_0+j+1}, with $|I_{j+1}| \geq d_2/(48\,t_{j_0+j+1})$.

The I_j are nested nonempty closed bounded subintervals of I_0, so by the nested interval theorem (equivalently, by compactness of I_0) the intersection $\bigcap_j I_j$ is nonempty. Fix $\gamma^* \in \bigcap_j I_j$; it is safe for every t_i simultaneously. By (3) and the criterion following it, $(kA_{\gamma^*,c}) \cap B = \emptyset$. As $A_{\gamma^*,c}$ is a (d_1, d_2)-sequence, this proves $r_k(d_1, d_2) \leq 192d_2$. □

This establishes part (1) of Theorem A; the rest of the paper proves part (2).

Remark 2.3. Theorem 2.1 is stated and proved here for all $k \geq 2$; the formalization deliberately carries the problem's hypothesis $k \geq 3$ throughout (see §6.4).

3. Non-existence for $d_2 \le k$: reduction to Lemma 1.4

Throughout this section $k \ge 3$ and $d_2 \le k$.

Glossary of terms coined in this section. Several constructions below are given short names; they are collected here with their formal content. (The terms of §4 — *frame*, *budget*, *staircase*, *λ-lift*, *hard core* — are defined where they arise, in §§4.1 and 4.3.)

Term	Formal content
tail-covering	$kA \supseteq \{x \ge X_0 : x \equiv \rho (\bmod m)\}$ for some m, ρ, X_0
tail alphabet G_∞	the gap values occurring infinitely often in (g_n)
gap word	the sequence (g_n) of consecutive differences of A
antidiagonal width $W_2(\sigma)$	$\max\{q_i + q_j : i + j = \sigma\} - \min\{q_i + q_j : i + j = \sigma\}$
fine slot	a summand index free to take one of two values, $n_t - 1$ or n_t
coarse dial	the last summand p_n, whose steps are $\le M$

3.1. The certificate.

Lemma 3.1 (Certificate)**.** *Fix any enumeration $(m_i, \rho_i)_{i \ge 1}$ of all pairs $\{(m, \rho) : m \ge 1,\ 0 \le \rho < m\}$ in which each pair occurs infinitely often. Let $R : \mathbb{N} \to \mathbb{N}$ be an* arbitrary *prescribed ratio sequence (no monotonicity is assumed). Set $b_1 := 1$, and for $i \ge 1$ let*

$$b_{i+1} \;:=\; \textit{the least integer } > \max\big(R(i)\, b_i,\ b_i,\ i\big) \textit{ that is } \equiv \rho_{i+1} \pmod{m_{i+1}}.$$

Then $B = \{b_1 < b_2 < \cdots\}$ satisfies $b_{i+1} \ge R(i)\, b_i$ for every i. If moreover every (d_1, d_2)-sequence A is tail-covering, then this single B defeats them all: $(kA) \cap B \ne \emptyset$ for every (d_1, d_2)-sequence A.

Proof. Each b_{i+1} exists (an arithmetic progression modulo m_{i+1} meets every half-line), and by construction $b_{i+1} > b_i$ and $b_{i+1} > R(i)b_i$, so B is strictly increasing, positive, and lacunary with the prescribed ratios. Note that R is arbitrary: no monotonicity is used, so a prescribed sequence need not be replaced by its nondecreasing envelope.

Let A be a (d_1, d_2)-sequence and pick (m, ρ, X_0) with $kA \supseteq \{x \ge X_0 : x \equiv \rho \pmod m\}$. Choose an index $i \ge 2$ with $(m_i, \rho_i) = (m, \rho)$ and $b_i \ge X_0$; this is possible because the pair (m, ρ) recurs infinitely often and $b_i \ge i \to \infty$ (B being a strictly increasing sequence of positive integers). Then $b_i \equiv \rho \pmod m$ and $b_i \ge X_0$, so $b_i \in kA$. □

Remark 3.2 (One B for all parameters). The certificate B depends only on the prescribed ratio sequence R and the chosen enumeration of pairs — not on k, d_1 or d_2. Since every (d_1, d_2)-sequence with $d_2 \le k$ turns out to be tail-covering (Proposition 3.20), this *single* B simultaneously defeats every (d_1, d_2)-sequence for every $k \ge 3$ with $d_2 \le k$: a strengthening of Theorem A(2) beyond its statement. We also note that the strong form of [BHJ97] prescribes an *increasing* sequence of ratio bounds, whereas (r_i) here is arbitrary.

So it suffices to prove:

every (d_1, d_2)-sequence with $d_2 \le k$ is tail-covering.

The nontrivial modulus in the definition of tail-covering is essential: kA need not be cofinite, so one cannot always hope for "all large integers" ($m = 1$), and the modulus can even exceed d_2. For instance, gaps $3, 5, 3, 5, \ldots$ with $k = 5$ give exactly the six classes $5a_1 + \{0, 1, 3, 4, 6, 7\}$ (mod 8).

3.2. **Elementary reductions.**

Lemma 3.3 (Tail reduction)**.** *Let T be an index past which every gap of A lies in G_∞. If the k-fold sums of the tail $A \cap [a_T, \infty)$ are tail-covering, then so is A, since kA contains them.* □

We pass to such a tail silently from now on.

Lemma 3.4 (One letter)**.** *If $G_\infty = \{\delta\}$ then the tail of A is an arithmetic progression and $kA \supseteq \{ka_T + \delta j : j \geq 0\}$, so A is tail-covering with $m = \delta$.* □

Lemma 3.5 (Rescaling)**.** *If $g := \gcd(G_\infty) > 1$, write the tail of A as $a_T + gA'$, where A' has gap alphabet G_∞/g, of gcd 1 and maximum $\leq d_2/g$. If the k-fold sums of A' are tail-covering modulo m', then kA is tail-covering modulo gm'. Hence we may and do assume $\gcd(G_\infty) = 1$.* □

Lemma 3.6 (Eventually periodic)**.** *If the gap word of A is eventually periodic, then A is tail-covering.*

Proof. The tail of A is a full pattern $(C + P\mathbb{Z}) \cap [T', \infty)$ for a finite set C and period-sum P. For each k-element multiset of C with sum σ, the sumset kA contains the entire class tail $\sigma + P\mathbb{Z}$ beyond a point; so A is tail-covering. □

3.3. **The two-letter core.** For orientation, since this subsection borrows from combinatorics on words: a binary word is *balanced* if any two equal-length windows have 1-counts differing by at most one, and the aperiodic balanced words are exactly the *Sturmian* (or *mechanical*) words — those with counting function $q_n = \lfloor n\alpha + \beta \rfloor$ for an irrational slope α, the cutting sequences of lines of irrational slope. They matter here because they are the gap words with the least usable irregularity: regular enough to defeat the width arguments below, without ever becoming periodic. They are the delicate case of this section.

Assume now $G_\infty = \{\delta < \delta'\}$ with $\gcd(\delta, \delta') = 1$ (Lemma 3.5), both letters occurring infinitely often, and set $e := \delta' - \delta \geq 1$. Since every gap lies in $[d_1, d_2]$ and $d_2 \leq k$, the larger letter satisfies

$$\delta' \;=\; \delta + e \;\leq\; d_2 \;\leq\; k. \tag{5}$$

(The tail alphabet need not attain d_2; only the inequality (5) is used.) Write the gaps as $g_n = \delta + eh_n$ with $h_n \in \{0, 1\}$, and let $q_n := h_1 + \cdots + h_n$ be the counting function of the binary *gap word* h. By (2), every $x \in kA - ka_1$ has the form

$$x = \delta s + ew, \qquad s = \sum_t i_t, \qquad w = \sum_t q_{i_t},$$

so that w ranges over $W_s := \{\sum_t q_{i_t} : \sum_t i_t = s\}$.

Lemma 3.7 (Interval property)**.** *For each s, the set W_s is a full integer interval $[w^-(s), w^+(s)]$, and $w^\pm(s+1) - w^\pm(s) \in \{0, 1\}$.*

Proof. The set of k-multisets of nonnegative integers with fixed sum s is connected under the moves $(i_a, i_b) \mapsto (i_a - 1, i_b + 1)$, and each such move changes $\sum_t q_{i_t}$ by $h_{i_b+1} - h_{i_a} \in \{-1, 0, 1\}$. A path from a minimizer to a maximizer therefore visits a set of integer values with no gap of size ≥ 2, so W_s is an interval. The increment claim follows by comparing multisets for s and $s + 1$ differing in a single index. □

Lemma 3.8 (Sweep)**.** *If $w^+(s) - w^-(s) \geq d_2 - 1$ for all $s \geq S_0$, then $kA - ka_1$ contains all sufficiently large integers; in particular A is tail-covering with $m = 1$.*

The mechanism is the picture in Figure 2: as s grows the window $[w^-(s), w^+(s)]$ of reachable values widens, while the target $w(s)$ needed to represent a fixed large x descends; because a widening band cannot be stepped over by a descending target, some admissible s lands the target inside the window, and that s realizes x.

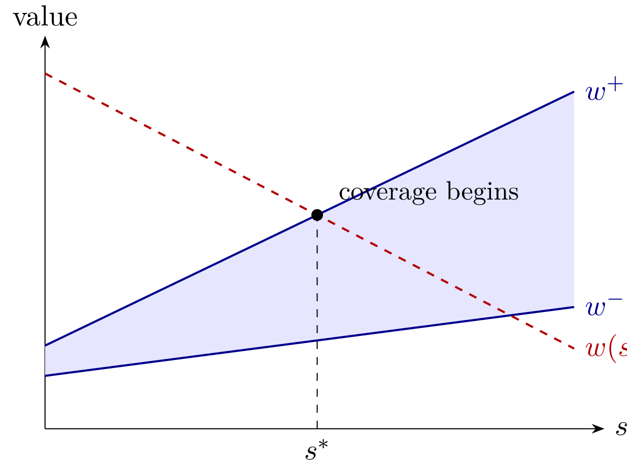

FIGURE 2. (Lemma 3.8, schematic.) The window $[w^-(s), w^+(s)]$ of reachable values (solid boundaries) widens with s, while the target $w(s)$ for representing a fixed large x (dashed) descends into it; past the crossing s^* every large value is reachable.

Proof. Let x be large. Representations $x = \delta s + ew$ force $\delta s \equiv x \pmod{e}$, a single residue class since $\gcd(\delta, e) = 1$, so the admissible s form an arithmetic progression of common difference e, and for each such s the value $w(s) := (x - \delta s)/e$ increases by exactly δ when s decreases by e. The windows $[w^-(s), w^+(s)]$ are nondecreasing in s with increments in $[0, e]$ per admissible step (Lemma 3.7). For the largest admissible s with $w(s) \geq 0$ we have $w(s) < \delta$; and $\delta \leq w^+(s)$ for all large s, since the letter 1 occurs infinitely often and hence $w^+(s) \geq q_s \to \infty$ — enlarging S_0 if necessary, we assume $w^+(s) \geq \delta$ for all $s \geq S_0$. At the other end, for the smallest admissible $s \geq S_0$ we have $w(s) = (x - \delta s)/e \geq (x - \delta(S_0 + e))/e$, which exceeds $w^+(s) \leq s$ once x is large (explicitly, once $x > \delta(S_0 + e) + e(S_0 + e)$); so $w(s) > w^+(s)$ there.

Suppose no admissible $s \geq S_0$ had $w(s) \in [w^-(s), w^+(s)]$. Since $w(\cdot)$ decreases along the admissible progression while the windows increase, there is then a crossing pair s', $s' + e$ with $w(s' + e) < w^-(s' + e)$ and $w(s') > w^+(s')$. We bound s' from below explicitly. The counting function satisfies $q_i \leq i$, so for any k indices with $\sum_t i_t = s$ we have $\sum_t q_{i_t} \leq s$; hence $w^+(s) \leq s$ for every s. At the index $s' + e$ the inequality has reversed, so

$$\frac{x - \delta(s' + e)}{e} \;=\; w(s' + e) \;\leq\; w^-(s' + e) \;\leq\; w^+(s' + e) \;\leq\; s' + e,$$

which rearranges to $s' + e \geq x/(\delta + e)$, that is

$$s' \;\geq\; \frac{x}{\delta + e} - e.$$

In particular $s' \geq S_0$ as soon as $x \geq (\delta + e)(S_0 + e)$, which we assume. But then

$$w^+(s') \;<\; w(s') \;=\; w(s' + e) + \delta \;\leq\; w^-(s' + e) - 1 + \delta \;\leq\; w^-(s') + e - 1 + \delta,$$

that is, $w^+(s') - w^-(s') \leq (\delta + e) - 2 = \delta' - 2 \leq d_2 - 2$, contradicting the hypothesis (which is where width $\geq d_2 - 1 \geq \delta' - 1$ is used). So some admissible $s \geq S_0$ has $w(s) \in [w^-(s), w^+(s)]$, and by Lemma 3.7 that value is attained. $\square$

For $\sigma \geq 0$ define the *two-index antidiagonal width*

$$W_2(\sigma) \;:=\; \max\{q_i + q_j : i + j = \sigma\} - \min\{q_i + q_j : i + j = \sigma\}.$$

Lemma 3.9 (Width dichotomy)**.**

(a) *If* $W_2(\sigma_0) \geq 2$ *for some* σ_0*, and either* k *is odd or* $d_2 \leq k-1$*, then* $w^+(s) - w^-(s) \geq d_2 - 1$ *for all sufficiently large* s*.*

(b) *If* $W_2(\sigma) \leq 1$ *for every* σ*, then the word* h *is* balanced.

Proof. (a) Disjoint index-pairs placed on antidiagonals contribute widths additively. For k odd, $(k-1)/2$ pairs at σ_0 plus one leftover index give $w^+(s) - w^-(s) \geq k-1 \geq d_2 - 1$ for all $s \geq \frac{k-1}{2}\sigma_0$, using exactly k indices. For k even and $d_2 \leq k-1$, $(k/2-1)$ pairs at σ_0 plus one free pair at $\tau := s - (k/2-1)\sigma_0$ give width $\geq (k-2) + W_2(\tau) \geq k-2 \geq d_2 - 1$.

(b) For any equal-length windows $(i, i']$ and $(j', j]$ one has $i + j = i' + j'$, whence $|(q_{i'} - q_i) - (q_j - q_{j'})| = |q_{i'} + q_{j'} - q_i - q_j| \leq 1$: the word h is balanced. □

The dichotomy routes as follows. In case (a), Lemma 3.8 gives tail-covering. In case (b), Proposition 3.11 classifies h: either ultimately periodic (Lemma 3.6), or, after passing to a tail, mechanical with irrational slope (Lemma 3.14); either way A is tail-covering.

The remaining two-letter configuration — k even at the boundary $d_2 = k$ — is the delicate one, and we route it at top level (independently of any width hypothesis):

Proposition 3.10 (Boundary routing; k even, $d_2 = k$)**.** *Let* k *be even and* $d_2 = k$*. Then* A *is tail-covering.*

Proof. At least one of three possibilities holds (they can overlap; we route by the first that applies), and each closes the case. If the gap word is eventually periodic, Lemma 3.6 applies. If some tail of the gap word is balanced, that tail (Lemma 3.3) is ultimately periodic (Lemma 3.6) or Sturmian (Lemma 3.14) by Proposition 3.11, and either gives tail-covering with no width argument. And if *every* tail of the gap word is unbalanced, Lemma 3.12 supplies width $\geq k-1$ for all large s, whence Lemma 3.8 gives tail-covering. □

The parity split in (a) is essential: when k is even, $\lfloor k/2 \rfloor$ pairs plus a leftover index would consume $k+1$ indices.

This dichotomy is the classical Morse–Hedlund theorem [MH40] (see also [Lo02, Ch. 2]), and a reader content to cite it may pass directly to Lemma 3.14. We nonetheless give a proof, for two reasons: we need only one direction of it, and the argument below is the one our formalization carries out — it is elementary, and its only analytic input is Dirichlet's approximation theorem. No density, equidistribution or three-distance statement is used anywhere.

Proposition 3.11 (Balanced classification)**.** *Let* $h : \mathbb{N} \to \{0,1\}$ *be balanced, with* $q_n := h_1 + \cdots + h_n$ *and* $q_0 := 0$*. Then* h *is either eventually periodic, or — after passing to a tail — satisfies* $q_n = \lfloor n\alpha + \beta \rfloor$ for every $n \geq 1$*, for some irrational* $\alpha \in (0,1)$ *and some* $\beta \in \mathbb{R}$*.*

Proof. Step 1: near-additivity. For all $n, m \geq 0$,

$$|q_{n+m} - q_n - q_m| \;\leq\; 1. \tag{6}$$

Indeed $q_{n+m} - q_n$ counts the letters 1 in the window $h_{n+1} \cdots h_{n+m}$, of length m, while q_m counts them in $h_1 \cdots h_m$, also of length m; balancedness bounds the difference of these two counts by 1.

Step 2: the slope exists, and $|q_n - n\alpha| \leq 1$*.* By (6) the sequence $a_n := q_n + 1$ is subadditive, $a_{n+m} \leq a_n + a_m$, and the sequence $b_n := q_n - 1$ is superadditive, $b_{n+m} \geq b_n + b_m$. By Fekete's lemma both a_n/n and b_n/n converge, to $\inf_{n\geq 1} a_n/n$ and $\sup_{n \geq 1} b_n/n$ respectively. Since a_n/n and b_n/n differ by $2/n \to 0$, the two limits coincide; call the common value α. Then $q_n/n \to \alpha$, and

$$\alpha \;=\; \sup_{n\geq 1} \frac{q_n - 1}{n} \;=\; \inf_{n \geq 1} \frac{q_n + 1}{n},$$

so that for every $n \geq 1$ we have $(q_n - 1)/n \leq \alpha \leq (q_n + 1)/n$, i.e.

$$|D_n| \ \leq \ 1, \qquad D_n := q_n - n\alpha \qquad (n \geq 0), \tag{7}$$

with $D_0 = 0$. Also $\alpha \in [0, 1]$, since $0 \leq q_n \leq n$.

Step 3: the oscillation bound. For all $n \geq 0$ and $\Delta \geq 1$,

$$\left|D_{n+\Delta} - D_n\right| \ \leq \ 1. \tag{8}$$

Fix Δ and write $W(j) := q_{j+\Delta} - q_j$ for the number of 1s in the length-Δ window starting at position $j + 1$. Balancedness gives $|W(j) - W(n)| \leq 1$ for all j. Summing over the t consecutive disjoint blocks starting at $0, \Delta, \ldots, (t-1)\Delta$ telescopes:

$$q_{t\Delta} \ = \ \sum_{i=0}^{t-1} W(i\Delta) \ \in \ \left[\, t\,(W(n) - 1),\ t\,(W(n) + 1)\,\right].$$

Dividing by $t\Delta$ and letting $t \to \infty$, where $q_{t\Delta}/(t\Delta) \to \alpha$ by Step 2, gives $(W(n) - 1)/\Delta \leq \alpha \leq (W(n) + 1)/\Delta$, i.e. $|W(n) - \Delta\alpha| \leq 1$. Since $D_{n+\Delta} - D_n = W(n) - \Delta\alpha$, this is (8). (Taking $n = 0$ recovers (7).)

Step 4: α rational $\Rightarrow$ h eventually periodic. Let $\alpha = p/q$ in lowest terms and put $c_n := q\,q_n - p\,n = qD_n$, an integer with $|c_n| \leq q$ by (7). With $w_n := q_{n+q} - q_n$ the length-q window counts,

$$c_{n+q} - c_n \ = \ q\,(w_n - p).$$

Balancedness confines the w_n to at most two consecutive values, and p is among them: if $w_n \geq p+1$ for every n then $c_{n+q} - c_n \geq q$ for every n, so c would increase without bound along each residue class modulo q, contradicting $|c_n| \leq q$; symmetrically, $w_n \leq p-1$ for every n would drive c to $-\infty$. Hence either $w_n \in \{p, p+1\}$ for all n, or $w_n \in \{p-1, p\}$ for all n.

In the first case $c_{n+q} - c_n \in \{0, q\}$, so along each residue class modulo q the sequence c is nondecreasing; being confined to $[-q, q]$, it is eventually constant on that class. In the second case c is nonincreasing, with the same conclusion. Either way $w_n = p$ for all large n, i.e. $q_{n+q} = q_n + p$ for all large n, whence

$$h_{n+q+1} \ = \ q_{n+q+1} - q_{n+q} \ = \ (q_{n+1} + p) - (q_n + p) \ = \ h_{n+1}$$

for all large n: the word h is eventually periodic, of period q.

Step 5: α irrational $\Rightarrow$ h is mechanical on a tail. Since α is irrational, $\alpha \notin \{0, 1\}$, so $\alpha \in (0, 1)$. Put

$$\beta \ := \ \sup_{n \geq 0} D_n,$$

which is finite by (7). By the oscillation bound (8), $D_n \geq \beta - 1$ for every n: indeed for any n and any m, $D_m - D_n \leq 1$, and taking the supremum over m gives $\beta - D_n \leq 1$.

Now observe that, for a given n,

$$q_n = \lfloor n\alpha + \beta \rfloor \iff n\alpha + \beta - 1 \ < \ q_n \ \leq \ n\alpha + \beta \iff \beta - 1 \ < \ D_n \ \leq \ \beta.$$

The right-hand inequality $D_n \leq \beta$ holds by the definition of β, and the left-hand one holds in the weak form $D_n \geq \beta - 1$ by the previous paragraph. So the identity can fail only when $D_n = \beta - 1$ exactly — equivalently, when $n\alpha + \beta = q_n + 1$ is an integer, i.e. $\{n\alpha + \beta\} = 0$. If two indices $n_1 \neq n_2$ both had $n_i\alpha + \beta \in \mathbb{Z}$, then $(n_1 - n_2)\alpha \in \mathbb{Z}$, which is impossible for irrational α. So there is *at most one* exceptional index; call it n_0 if it exists, and set $n_0 := 0$ otherwise. Only indices $n \geq 1$ are used below, so no assertion at $n = 0$ is required. Then

$$q_n \ = \ \lfloor n\alpha + \beta \rfloor \qquad \text{for every } n \geq 1 \text{ with } n \neq n_0.$$

Finally we pass to the tail beyond n_0 (Lemma 3.3). The tail word $h'_j := h_{n_0+j}$ has counting function $q'_j = q_{n_0+j} - q_{n_0}$, and for every $j \geq 1$ (so that $n_0 + j \neq n_0$),

$$q'_j \;=\; \lfloor (n_0+j)\alpha + \beta \rfloor - q_{n_0} \;=\; \lfloor j\alpha + (n_0\alpha + \beta - q_{n_0}) \rfloor \;=\; \lfloor j\alpha + \beta' \rfloor, \qquad \beta' := \beta - D_{n_0},$$

using that q_{n_0} is an integer. So the tail is mechanical with slope α and intercept β' for every $j \geq 1$. (Nothing is claimed at $j = 0$, where $q'_0 = 0$ by definition while $\lfloor \beta' \rfloor$ may be 1. Only indices $j \geq 1$ are used in Lemma 3.14, whose summand indices i_t are all ≥ 1.) $\square$

Everything above rests on the near-additivity (6) and the discrepancy bound (7), not on any density or equidistribution statement. Similarly, the uniform syndeticity of Lemma 3.13 below is obtained from a single Dirichlet approximation (a bounded circle walk), not from the three-distance theorem or any equidistribution input — at the cost of a worse but explicit constant. This is the route carried out in the formalization (`disc_osc_le_one`, `mechanical_tail`, `walk_enters`).

Lemma 3.12 (Boundary width; k even, $k \geq 4$)**.** *Let h use both letters infinitely often, let h be not eventually periodic, and suppose every tail of h is unbalanced. Then $w^+(s) - w^-(s) \geq k-1$ for all sufficiently large s. (No antidiagonal-width hypothesis is needed: the proof derives $W_2(\sigma) \geq 2$ for infinitely many σ from unbalancedness.)*

Proof. First we check that unbalancedness of *every* tail does give $W_2(\sigma) \geq 2$ for infinitely many σ. Suppose not, so that $W_2(\sigma) \leq 1$ for every $\sigma \geq \sigma^*$. Let $(i, i']$ and $(j', j]$ be any two equal-length windows lying beyond σ^*; then $i + j = i' + j'$, so $W_2(i+j) \leq 1$ gives $|q_{i'} + q_{j'} - q_i - q_j| \leq 1$, i.e.

$$\big|(q_{i'} - q_i) - (q_j - q_{j'})\big| \;\leq\; 1.$$

That is precisely the statement that the tail of h beyond σ^* is balanced, contradicting the hypothesis that every tail is unbalanced. Hence $W_2(\sigma) \geq 2$ for infinitely many σ.

Fix two of them, $\sigma_0 < \sigma_1$, and put $\Delta := \sigma_1 - \sigma_0$. For large s consider two configuration families, each using exactly k indices:

(1) $(k/2 - 1)$ extremal pairs at σ_0, plus one free pair at $\tau := s - (k/2-1)\sigma_0$, giving width $\geq (k-2) + W_2(\tau)$;
(2) $(k/2-2)$ extremal pairs at σ_0, one extremal pair at σ_1, plus one free pair at $\tau - \Delta$, giving width $\geq (k-2) + W_2(\tau - \Delta)$. (Here $k/2 - 2 \geq 0$ uses $k \geq 4$.)

If $w^+(s) - w^-(s) \leq k-2$, both force $W_2(\tau) = W_2(\tau - \Delta) = 0$. But $W_2(\tau) = 0$ means $q_i + q_{\tau - i}$ is constant in i, i.e. $h_{i+1} = h_{\tau - i}$ for $0 \leq i \leq \tau - 1$: the prefix $h[1..\tau]$ is a palindrome. Likewise $h[1..\tau-\Delta]$ is a palindrome. A palindromic prefix of a palindrome is also its suffix, so $h[1..\tau-\Delta]$ is a border of $h[1..\tau]$, and hence $h[1..\tau]$ has period Δ: this is the classical border–period duality (a word w with a border of length ℓ has period $|w| - \ell$, since the border condition $w_i = w_{i+|w|-\ell}$ holds for all $i \leq \ell$; see [Lo02]). If the width bound failed for infinitely many s, then, Δ being fixed while $\tau(s)$ strictly increases, we obtain prefixes $h[1..\tau]$ of period Δ with $\tau \to \infty$. For any fixed position n, choosing such a prefix of length $\tau > n + \Delta$ gives $h_n = h_{n+\Delta}$ from its period relation; as n was arbitrary, the entire infinite word h has period Δ, contradicting non-periodicity. $\square$

Lemma 3.13 (Uniform hitting for irrational rotation)**.** *Let $\alpha \in (0,1)$ be irrational and $\beta \in \mathbb{R}$. For every arc $J = (u, v) \subseteq (0,1)$ with $\varepsilon := v - u > 0$ there is an integer $L = L(\alpha, \varepsilon)$, depending only on α and ε — not on the position of J, nor on β — such that every block of L consecutive integers $\{N+1, \ldots, N+L\}$ contains some n with $\{n\alpha + \beta\} \in J$.*

Proof. An explicit *circle walk*, using only Dirichlet's approximation theorem. Choose an integer $Q > 2/\varepsilon$. By Dirichlet's theorem there are integers $1 \leq n_0 \leq Q$ and m with $|n_0\alpha - m| < 1/Q$. Set

$$\theta \;:=\; \|n_0\alpha\| \;=\; |n_0\alpha - m| \;\in\; (0,\; \tfrac{\varepsilon}{2}),$$

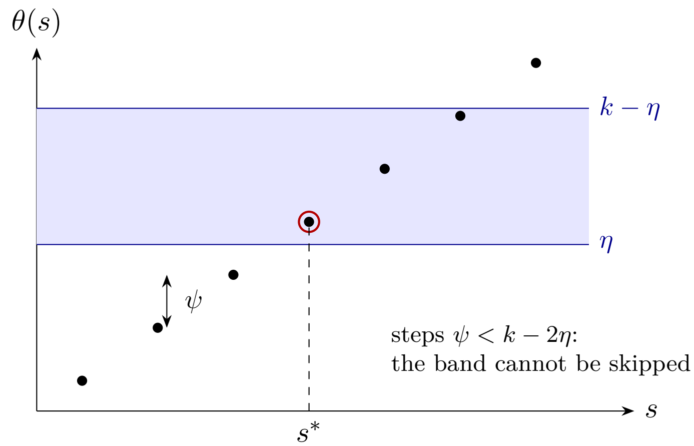

FIGURE 3. (Lemma 3.14, Step 3, schematic.) Along the congruence-compatible progression, the depth $\theta(s)$ climbs in equal steps $\psi = e\alpha + \delta$; since ψ is strictly smaller than the sub-window's height $k - 2\eta$, some s^* lands inside the band, and Step 2 realizes the corresponding target.

nonzero because α is irrational; one step of n_0 in the index moves $\{n\alpha + \beta\}$ around the circle by exactly $\pm\theta$, the sign being that of $n_0\alpha - m$, the same at every step. Fix any $N \geq 0$ and consider the $\lceil 1/\theta \rceil + 1$ indices $n_j := N + (j+1)\, n_0$ ($j = 0, \ldots, \lceil 1/\theta \rceil$), all lying in $\{N+1, \ldots, N+L\}$ with $L := n_0(\lceil 1/\theta \rceil + 1) \leq Q(\lceil 1/\theta \rceil + 1)$. The points $x_j := \{n_j\alpha + \beta\}$ satisfy $x_{j+1} = \{x_j + \theta\}$ after orienting the circle so the fixed step direction is positive; join each x_j to x_{j+1} by the positively oriented arc of length θ. Over the $\lceil 1/\theta \rceil$ steps the total oriented length is at least 1, so these arcs cover the whole circle. If no x_j lay in J, some single step would have to jump across all of J; but each step has length $\theta < \varepsilon/2 < \varepsilon = |J|$, which is impossible. Hence some $x_j \in J$. The constant L depends only on α and ε — not on where J sits, nor on β — the argument using only the step length θ and its direction. ☐

Lemma 3.14 (Sturmian case)**.** *Let the two-letter tail of §3.3 have $q_n = \lfloor n\alpha + \beta \rfloor$ for an irrational $\alpha \in (0, 1)$ and some $\beta \in \mathbb{R}$. Then $kA - ka_1$ contains all sufficiently large integers, so A is tail-covering with $m = 1$.*

Proof. We keep the notation of §3.3: $e = \delta' - \delta \geq 1$, $\gcd(\delta, e) = 1$, and $\delta' = \delta + e \leq k$ by (5). Every $x \in kA - ka_1$ has the form $x = \delta s + ew$ with $s = \sum_t i_t$ (all $i_t \geq 1$, as we work in the tail) and $w = \sum_t q_{i_t} \in W_s$.

Overview. The real window $(X_s - k,\ X_s]$ of (9) carries W_s (Step 0). The uniform hitting lemma for the irrational rotation by α (Lemma 3.13) lets us fill an interior sub-window: every integer of $[\eta, k - \eta]$-depth below X_s is realized as some $\sum_t q_{i_t}$ (Step 2). Finally a congruence ladder is walked across that sub-window to represent an arbitrary large x, with explicit thresholds making the chosen indices and coefficient nonnegative (Step 3, Figure 3).

Step 0: the window carrying W_s. From $q_n = n\alpha + \beta - \{n\alpha + \beta\}$,

$$\sum_t q_{i_t} \;=\; \Big(\alpha \sum_t i_t + k\beta\Big) - \sum_t \{i_t\alpha + \beta\} \;=\; X_s - \sum_t \{i_t\alpha + \beta\}, \qquad X_s := s\alpha + k\beta.$$

Each $\{i_t\alpha + \beta\} \in [0, 1)$, so $\sum_t \{i_t\alpha + \beta\} \in [0, k)$ and hence

$$W_s \;\subseteq\; (X_s - k,\ X_s] \cap \mathbb{Z}. \tag{9}$$

Step 1 (uniform hitting). This is Lemma 3.13: for every arc $J \subseteq (0,1)$ of length ε, every block of $L = L(\alpha, \varepsilon)$ consecutive indices contains an n with $\{n\alpha + \beta\} \in J$. Fix this L for the specific arc chosen in Step 2.

Step 2: the sub-window claim. Fix $\eta \in (0, \frac{1}{2})$. We claim there is $S_0 = S_0(\eta, \alpha, \beta, k)$ such that for every $s \geq S_0$, the set W_s contains every integer w for which

$$\theta \;:=\; X_s - w \;\in\; [\eta,\; k - \eta]. \tag{10}$$

That is: within the real window $(X_s - k, X_s]$ of (9), every integer point whose distance θ below X_s lies in $[\eta, k - \eta]$ is attained.

To see this, fix $s \geq S_0$ and an integer w with $\theta \in [\eta, k - \eta]$. It suffices to produce pairwise distinct indices $i_1, \ldots, i_k \geq 1$ with

$$\sum_{t=1}^{k} i_t = s \qquad \text{and} \qquad \sum_{t=1}^{k} \{i_t\alpha + \beta\} = \theta \;\text{ exactly,} \tag{11}$$

since then Step 0 gives $\sum_t q_{i_t} = X_s - \theta = w$, i.e. $w \in W_s$.

Set $\kappa := \eta/4$ and consider the open real interval

$$\mathcal{I} \;:=\; \big(\max(0, \theta - 1) + \kappa,\; \min(k-1, \theta) - \kappa\big).$$

It is nonempty with $|\mathcal{I}| \geq \eta/2$, in each of the three regimes of θ: if $\eta \leq \theta \leq 1$ then $|\mathcal{I}| = \theta - 2\kappa \geq \eta/2$; if $1 < \theta \leq k-1$ then $|\mathcal{I}| = 1 - \eta/2 > \eta/2$; and if $k - 1 < \theta \leq k - \eta$ then $|\mathcal{I}| = (k - \theta) - 2\kappa \geq \eta/2$.

Pick any $T^* \in \mathcal{I}$ and set the target arc

$$J \;:=\; \Big(\tfrac{T^*}{k-1} - \tfrac{\kappa}{k-1},\; \tfrac{T^*}{k-1} + \tfrac{\kappa}{k-1}\Big) \;\subseteq\; (0,1),$$

of length $2\kappa/(k-1) > 0$; let $L = L\big(\alpha, 2\kappa/(k-1)\big)$ be its hitting constant from Lemma 3.13. The arc J moves with θ (through T^*), but its *length* does not, and Lemma 3.13 is uniform in the arc's position — so this single L serves every target w and every s. Choose the $k-1$ *free* indices $i_1 < \cdots < i_{k-1}$, all with $\{i_t\alpha + \beta\} \in J$, from the lowest available blocks: for $j = 1, \ldots, k-1$ the block $B_j := \{(j-1)L + 1, \ldots, jL\}$ contains, by Lemma 3.13, at least one index i_j with $\{i_j\alpha + \beta\} \in J$. These lie in disjoint blocks, so they are distinct, and

$$i_j \leq jL, \qquad \sum_{t=1}^{k-1} i_t \leq L\,\frac{k(k-1)}{2}, \qquad i_{k-1} \leq (k-1)L. \tag{12}$$

Define the threshold

$$S_0 \;:=\; 1 + \Big\lceil L\,(k-1)\Big(\frac{k}{2} + 1\Big)\Big\rceil \tag{13}$$

(the ceiling makes it an integer: L is an integer, but the factor $\frac{k}{2}$ is a half-integer when k is odd), and assume $s \geq S_0$. Set the forced last index $i_k := s - \sum_{t=1}^{k-1} i_t$. Then by (12) and (13),

$$i_k \;\geq\; s - L\frac{k(k-1)}{2} \;\geq\; S_0 - L\frac{k(k-1)}{2} \;=\; 1 + (k-1)L \;>\; (k-1)L \;\geq\; i_{k-1},$$

so $i_k \geq 1$ and $i_k > i_{k-1} > \cdots > i_1$: the indices $i_1, \ldots, i_k \geq 1$ are pairwise distinct with $\sum_t i_t = s$.

With these choices, writing $F := \sum_{t=1}^{k-1}\{i_t\alpha + \beta\}$,

$$F \;\in\; (T^* - \kappa,\; T^* + \kappa) \;\subseteq\; \big(\max(0, \theta - 1),\; \min(k-1, \theta)\big), \tag{14}$$

the last inclusion because $T^* \in \mathcal{I}$ sits at distance $> \kappa$ from each endpoint. Now put $\Theta := \sum_{t \leq k}\{i_t\alpha + \beta\} = F + \{i_k\alpha + \beta\}$. From (14) and $\{i_k\alpha + \beta\} \in [0,1)$,

$$\Theta > \max(0, \theta - 1) + 0 \geq \theta - 1, \qquad \Theta < \min(k-1, \theta) + 1 \leq \theta + 1,$$

so $\Theta \in (\theta-1, \theta+1)$. On the other hand

$$\Theta \;=\; \sum_{t\le k}(i_t\alpha+\beta) - \sum_{t\le k}\lfloor i_t\alpha+\beta\rfloor \;=\; X_s - \sum_{t\le k} q_{i_t} \;\in\; X_s - \mathbb{Z} \;=\; \theta+\mathbb{Z},$$

using $\sum_{t\le k}(i_t\alpha+\beta) = \alpha s + k\beta = X_s$ exactly and $w = X_s - \theta \in \mathbb{Z}$. The unique element of the coset $\theta+\mathbb{Z}$ lying in the open interval $(\theta-1, \theta+1)$ is θ itself, so $\Theta = \theta$. This proves (11) and hence Step 2.

Step 3: the ladder lands in the sub-window. Instantiate Step 2 with

$$\eta \;:=\; \tfrac{1}{4}\min(\alpha,\; 1-\alpha) \;\in\; (0, \tfrac{1}{8}], \tag{15}$$

and fix the resulting S_0. Let x be a large integer. Because $\gcd(\delta, e) = 1$, the congruence $\delta s \equiv x \pmod{e}$ determines a single residue class; call its members the *congruence-compatible* s, an arithmetic progression of common difference e. For each such s set $w(s) := (x-\delta s)/e \in \mathbb{Z}$; the representation $x = \delta s + e\,w(s)$ uses nonnegative integers exactly when $s \ge 0$ and $w(s) \ge 0$, which we confirm at the crossing index below. Define

$$\theta(s) \;:=\; X_s - w(s) \;=\; (s\alpha + k\beta) - \frac{x-\delta s}{e}.$$

As s advances by one congruence-compatible step ($s \mapsto s+e$), X_s increases by $e\alpha$ while $w(s)$ decreases by δ, so

$$\theta(s+e) - \theta(s) \;=\; e\alpha + \delta \;=:\; \psi \;>\; 0. \tag{16}$$

Thus θ is, along the congruence-compatible progression, a strictly increasing arithmetic sequence of common difference exactly ψ.

The sub-window (10) is the real interval $\theta \in [\eta, k-\eta]$, of length $k - 2\eta$. Using $\delta + e = \delta' \le k$ and (15),

$$\psi \;=\; \delta + e\alpha \;=\; (\delta+e) - e(1-\alpha) \;=\; \delta' - e(1-\alpha) \;\le\; k - (1-\alpha),$$

since $e \ge 1$ and $1-\alpha > 0$. As $2\eta \le \frac{1}{2}(1-\alpha) < 1-\alpha$, we conclude

$$\psi \;\le\; k - (1-\alpha) \;<\; k - 2\eta, \tag{17}$$

i.e. the ladder step is *strictly* shorter than the sub-window.

A strictly increasing arithmetic sequence of common difference ψ cannot skip over a real interval of length $> \psi$: if $\theta(s) < \eta$ and $\theta(s+e) > k-\eta$ for two consecutive congruence-compatible terms, then $\psi > (k-\eta) - \eta = k - 2\eta$, contradicting (17). As s runs through the congruence-compatible progression, $\theta(s)$ increases without bound and is very negative for the smallest congruence-compatible s (where $w(s) \approx x/e$ dominates), so it crosses $[\eta, k-\eta]$; by the no-skip property some congruence-compatible s^* has $\theta(s^*) \in [\eta, k-\eta]$.

It remains to say what "x large" means, and we make the two thresholds explicit. Solving $\theta(s) = \theta$ for s gives $s\psi = x + e\theta - ek\beta$, so any s^* with $\theta(s^*) \in [\eta, k-\eta] \subseteq [0, k]$ satisfies

$$\frac{x - ek|\beta|}{\psi} \;\le\; s^* \;\le\; \frac{x + ek(1+|\beta|)}{\psi}. \tag{18}$$

Two conditions are needed.

- $s^* \ge S_0$, *so that Step 2 applies.* By the left inequality of (18) and $\psi \le k$, it suffices that
$$x \;\ge\; k\,S_0 + e\,k\,|\beta|.$$
- $w(s^*) \ge 0$, *so that the representation $x = \delta s^* + e\,w(s^*)$ uses a nonnegative w.* This says $\delta s^* \le x$. By the right inequality of (18) this holds as soon as $\delta\big(x + ek(1+|\beta|)\big) \le \psi x = (\delta + e\alpha)x$, i.e. as soon as
$$x \;\ge\; \frac{\delta\,k\,(1+|\beta|)}{\alpha},$$

the denominator being nonzero because $\alpha \in (0,1)$.

So the conclusion holds for every integer

$$x \;\geq\; X_0 \;:=\; 1 + \max\Big(k\,S_0 + e\,k\,|\beta|, \quad \frac{\delta\,k\,(1+|\beta|)}{\alpha} \Big),$$

an explicit threshold depending only on k, δ, e, α, β (through S_0, which was fixed in Step 2). Step 2 then gives $w(s^*) \in W_{s^*}$, whence

$$x \;=\; \delta s^* + e\,w(s^*) \;\in\; kA - ka_1.$$

As x was an arbitrary large integer, $kA - ka_1$ contains all large integers; in the original, unscaled coordinates this is a full congruence-class tail modulo g, so A is tail-covering. □

Example 3.15 (the Fibonacci tail, traced). Take $k = 3$, letters $\{\delta, \delta'\} = \{1, 2\}$ (so $\delta = e = 1$), and the Fibonacci gap word: $q_n = \lfloor n\alpha + \beta \rfloor$ with $\alpha = (3 - \sqrt{5})/2 \approx 0.382$ and $\beta = \alpha$. Step 3's parameters are $\eta = \frac{1}{4}\min(\alpha, 1-\alpha) = \alpha/4 \approx 0.095$ and ladder step $\psi = e\alpha + \delta = 1 + \alpha \approx 1.382$, comfortably below the sub-window height $k - 2\eta \approx 2.81$. Since $e = 1$, every s is congruence-compatible and $w(s) = x - s$. To represent $x = 100$: $\theta(s) = X_s - w(s) = (s+3)\alpha - 100 + s$ climbs by ψ per step and first enters $[\eta, 3-\eta]$ at $s^* = 72$, where $\theta(72) = 75\alpha - 28 \approx 0.647$. So $w = 28$, at depth ≈ 0.647 inside the window $(X_{72} - 3,\ X_{72}] \approx (25.65,\ 28.65]$; Step 2 then supplies three distinct indices $i_1 < i_2 < i_3$ with $i_1 + i_2 + i_3 = 72$ and $q_{i_1} + q_{i_2} + q_{i_3} = 28$, whence $100 = 1 \cdot 72 + 1 \cdot 28 \in 3A - 3a_1$. This is the configuration of Figure 3.

Theorem 3.16 (Two-letter core)**.** *Let $k \geq 3$ and $d_2 \leq k$. Every (d_1, d_2)-sequence whose tail alphabet has at most two letters is tail-covering. In particular, with Lemma 3.1, $r_k(d_1, d_1+1)$ does not exist whenever $d_1 + 1 \leq k$, in the strong form.*

Proof. Lemmas 3.4, 3.5, 3.6 and 3.9, Proposition 3.10 at the even boundary $d_2 = k$, and Lemma 3.14 exhaust the cases. □

3.4. **Three or more letters: widths are not the obstacle.**

Remark 3.17. The antidiagonal width W_2 of §3.3 was defined through the binary counting function q_n, and has no meaning for an alphabet of three or more letters. We do not attempt to generalize it, because the route below does not need it. Indeed Lemma 3.7 already *fails* for three or more letters: configuration moves change values by letter *differences*, which can exceed 1, so W_s is dense in its span but may have holes, and Lemma 3.8 does not apply. The slot lemma below bypasses the antidiagonal analysis entirely.

3.5. **The slot lemma.**

Lemma 3.18 (Slot lemma)**.** *Let A have tail alphabet G_∞ with $|G_\infty| \geq 3$, $\gcd(G_\infty) = 1$ (after Lemma 3.5), and $M := \max G_\infty \leq d_2$. If $k - 1 \geq m(G_\infty)$, then A is tail-covering.*

Proof. Let S realize $m(G_\infty)$: a multiset with $|S| = m \leq k-1$, drawn from G_∞, whose subset sums contain $\{c, c+1, \ldots, c+M-1\}$. Enumerate the elements of S as $\delta_1, \ldots, \delta_m$ (values in G_∞, with repetition). Because each value of G_∞ occurs infinitely often in the gap sequence, we may pick m distinct positions $n_1 < \cdots < n_m$ with $g_{n_t} = \delta_t$; distinctness is all that is needed, since repeated summand indices are permitted in kA.

Call t a *fine slot*: a summand index $i_t \in \{n_t - 1,\ n_t\}$, contributing $p_{n_t - 1}$ ("off") or $p_{n_t-1} + \delta_t$ ("on"). Over all on/off choices the m fine slots contribute

$$\text{base} + \text{subset-sums}(S) \;\supseteq\; \text{base} + [\,c,\ c+M-1\,], \qquad \text{base} := \sum_t p_{n_t - 1}.$$

Park the remaining $k-1-m$ summands at fixed indices, folding their contribution into base (repeated indices are allowed in kA). Together with the coarse dial below this accounts for all k summands,

$$\underbrace{m}_{\text{fine slots}} + \underbrace{(k-1-m)}_{\text{parked}} + \underbrace{1}_{\text{dial}} = k,$$

and the $k-1$ summands so far realize every value of an interval of M consecutive integers.

Two endpoint facts should be made explicit. First, each value of G_∞ recurs infinitely often, so the positions n_t may be taken as far out as we please; we take them all beyond the tail index T of Lemma 3.3, so that $n_t - 1 \geq T$ and the "off" choice p_{n_t-1} is a legitimate prefix sum of the tail.

Second, the coarse dial. In the tail, consecutive prefix sums differ by $g_{n+1} \in G_\infty$, hence by at least 1 and at most M; so $n \mapsto p_n$ is strictly increasing with steps of size at most M, and $p_n \to \infty$. Fix x and put $\phi(n) := x - \text{base} - p_n$, which is strictly decreasing with steps of size at most M and tends to $-\infty$. Take x large enough that $\phi(T) > c + M - 1$ *strictly* — possible because base, T and c are fixed while x grows. Then ϕ starts strictly above the target interval $[c,\ c+M-1]$ and eventually falls below it, so there is a *first* index $n^* > T$ with $\phi(n^*) \leq c+M-1$; note $n^* > T$, so $n^*-1 \geq T$ is a legitimate tail index. By minimality $\phi(n^*-1) > c+M-1$, and since ϕ decreases by at most M in one step,

$$\phi(n^*) \ > \ (c+M-1)-M \ = \ c-1, \qquad \text{i.e.} \qquad \phi(n^*) \ \geq \ c.$$

Hence $\phi(n^*) \in [c,\ c+M-1]$, an interval realized by the fine slots. (The dial index n^* may coincide with a fine-slot or parked index; that is harmless, since repetitions are allowed in kA.) This first-crossing argument is exactly where a run of M consecutive integers is needed: a run of length $M-1$ would not suffice, since the dial can step by M.

Therefore $x \in kA - ka_1$, and $kA - ka_1$ contains all large integers (in rescaled coordinates; a full class tail modulo g in general). □

The binding corner is $k = d_2 = M$ — the adversary uses the maximal gap infinitely often and k is as small as the hypothesis $d_2 \leq k$ allows. There the requirement $k-1 \geq m(G_\infty)$ becomes *exactly* $m(G_\infty) \leq M-1$, which is precisely Lemma 1.4. Granting it: $k-1 \geq d_2-1 \geq M-1 \geq m(G_\infty)$ for all $k \geq d_2$ (and for rescaled alphabets $M/g - 1 < d_2 - 1$ a fortiori).

For perspective we record a short self-contained bound in the same direction. It is not needed below, but it exhibits the mechanism. It is genuinely weaker than Lemma 1.4: besides the worse bound, its hypothesis can fail — for $G = \{6, 10, 15\}$, of gcd 1, no element is coprime to $\min G = 6$.

Lemma 3.19 (Base grid)**.** *Let G satisfy $|G| \geq 3$, $\gcd G = 1$, $\min G = a$, $\max G = M$, and suppose G contains an element c with $\gcd(a,c) = 1$. Then $m(G) \leq a + M - 1 \leq 2M - 3$.*

Proof. Take $a-1$ copies of c and $p := \lceil ((a-1)c + M - 1)/a \rceil$ copies of a. For $n \in [(a-1)c,\ pa]$, the unique $i \in [0, a-1]$ with $ic \equiv n \pmod a$ satisfies $ic \leq (a-1)c \leq n$ and $(n-ic)/a \in [0,p]$, so the subset sums cover that interval, whose length is $\geq M$. Since $p \leq M$, the size is $\leq a+M-1$; and $|G| \geq 3$ forces $a \leq M-2$. □

3.6. **Conclusion of the non-existence half.**

Proposition 3.20. *Modulo Lemma 1.4, every (d_1, d_2)-sequence with $d_2 \leq k$ and $k \geq 3$ is tail-covering.*

Proof. Tails with at most two letters are covered by Theorem 3.16. Tails with three or more letters are covered by Lemma 3.18 together with Lemma 1.4, which supplies $m(G_\infty) \leq \max(G_\infty) - 1 \leq d_2 - 1 \leq k-1$. □

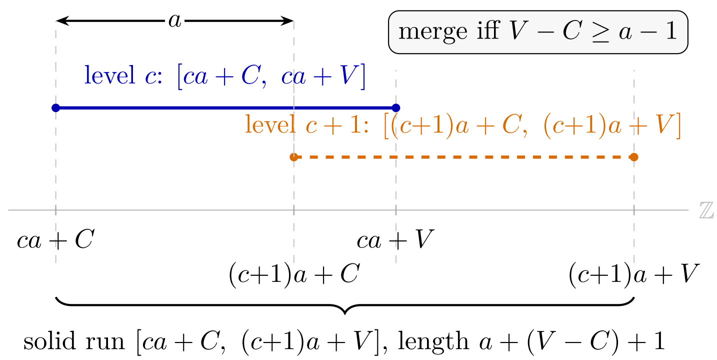

FIGURE 4. (Lemma 4.3, schematic.) Two consecutive frames — the intervals covered at levels c and $c + 1$ — are offset by a and merge into one solid run of consecutive integers precisely when $V - C \geq a - 1$.

With Lemma 3.1 this proves the non-existence half of Theorem A, in the strong form. It remains to prove Lemma 1.4.

4. The bounded subset-sum covering lemma

Before the details, the shape of the argument. We must cover M consecutive integers with a bounded multiset drawn from G. There is one mechanism (Figure 4): cover a short interval — a *frame* — with copies of the two largest generators, then slide and stack copies of the smallest generator a until the frames merge into one solid run. Everything reduces to a *three-generator hard core* $\{a, b, M\}$ (Lemmas 4.6–4.12). There the behaviour is governed by two arithmetic questions — whether $a \mid M$, and whether $a \mid b + M$ — together with the size of $\mu = M - a$. These invariants carve the hard core into six cases. Small a or small μ leave only finitely many triples to check by explicit certificate; the rest close by uniform symbolic constructions. The case map is Table 1, and the constant $M - 1$ is what the tightest case spends.

We restate the target.

Lemma 1.4 (restated). *For every finite $G \subset \mathbb{Z}_{\geq 1}$ with $|G| \geq 3$, $\gcd(G) = 1$ and $\max(G) = M$, there is a multiset S of elements of G with $|S| \leq M - 1$ whose subset sums contain M consecutive integers.*

If $a := \min G = 1$, take $M - 1$ copies of 1. So assume $a \geq 2$; the hard core below will force $a \geq 3$.

4.1. Machinery.

The constructions below all work by the same picture (Figure 4). Using a residue box of copies of b and M we cover one interval — a *frame* — at each "level" (number of summands used); adding one more copy of a shifts the whole frame right by a. Consecutive frames then abut into a single solid run of consecutive integers exactly when each frame is at least $a - 1$ long. The lemmas of this subsection make that quantitative.

Lemma 4.1 (Two-generator interval). *Let $\alpha < \beta$ be coprime positive integers, and let $x \geq \beta - 1$, $y \geq \alpha - 1$. Then, with $C := (\alpha - 1)(\beta - 1)$,*

$$\{\, i\alpha + j\beta \ :\ 0 \leq i \leq x,\ 0 \leq j \leq y \,\} \ \supseteq\ [\, C,\ \alpha x + \beta y - C\,].$$

Proof. If $\alpha = 1$ then $C = 0$; for $n \in [0, x + \beta y]$ take $j := \min(y, \lfloor n/\beta \rfloor)$, so that $0 \leq n - j\beta \leq \max(\beta - 1,\ n - y\beta) \leq x$.

Let $\alpha \geq 2$. *Base case $y = \alpha - 1$:* for $n \in [C,\ \alpha x + \beta(\alpha-1) - C]$ pick the unique $j \in [0, \alpha-1]$ with $j\beta \equiv n \pmod{\alpha}$. Then $n - j\beta \geq C - (\alpha-1)\beta = -(\alpha-1)$ and is $\equiv 0 \pmod{\alpha}$, hence ≥ 0; and $i := (n - j\beta)/\alpha \leq (\alpha x + (\alpha - 1 - j)\beta - C)/\alpha \leq x + (\alpha-1)/\alpha$, so $i \leq x$.

Step $y \to y+1$: the new part of the interval is $n \in (\alpha x + \beta y - C,\ \alpha x + \beta(y+1) - C]$. Then $n - \beta \in [C,\ \alpha x + \beta y - C]$ — the lower end because $\alpha x + \beta y \geq 2C + \beta$ for $x \geq \beta - 1$, $y \geq \alpha - 1$, $\alpha \geq 2$ — so by induction $n - \beta = i\alpha + j\beta$ with $i \leq x$, $j \leq y$, and thus $n = i\alpha + (j+1)\beta$. □

Lemma 4.2 (Frame lemma)**.** *Let g_1, g_2, ν be positive integers and $L \geq 1$. Suppose that for every residue $\rho \in \mathbb{Z}_\nu$ there is a pair (j_ρ, k_ρ) with $0 \leq j_\rho \leq Y$, $0 \leq k_\rho \leq Z$ and $j_\rho g_1 + k_\rho g_2 \equiv \rho \pmod{\nu}$. Put $S := \max_\rho(j_\rho g_1 + k_\rho g_2)$ and $x := \lceil (L - 1 + S)/\nu \rceil$. Then the multiset consisting of Y copies of g_1, Z copies of g_2 and x copies of ν — of size $Y + Z + x$ — has subset sums containing every integer of $[\,S,\ \nu x\,] \supseteq [\,S,\ S + L - 1\,]$.*

Proof. For $n \in [S, \nu x]$ let $\rho := n \bmod \nu$ and $r := j_\rho g_1 + k_\rho g_2 \leq S \leq n$. Then $t := (n - r)/\nu$ is a nonnegative integer $\leq x$, and taking j_ρ copies of g_1, k_ρ of g_2 and t of ν realizes n. □

In all uses below $\nu = a \in G$, $(g_1, g_2) = (b, M)$ and $L = M$; the *budget* of a box (Y, Z) is

$$\mathcal{B} \;=\; \left\lceil \frac{M - 1 + S}{a} \right\rceil + Y + Z,$$

with S the maximum, over residue classes, of the minimal representative height in the box. Standing notation. For a triple $G = \{a, b, M\}$ with $a < b < M$ and $\gcd(a, b) = 1$, set

$$e := b - a, \quad h := M - b, \quad \mu := M - a = e + h, \quad g := \gcd(e, \mu), \quad e' := e/g, \quad \mu' := \mu/g, \quad C' := (e'-1)(\mu'-1),$$

so that e', μ' are coprime with $\mu' \geq 2$ (as $h \geq 1$). Note that $\gcd(a, b) = 1$ forces $\gcd(a, g) = 1$: a common prime of g and a would divide e and a, hence b. This notation persists through the rest of the section, and is collected together with the hard-core symbols in the table of §4.3.

The level decomposition. Fix a multiset (x, y, z) of copies of a, b, M with $y \geq \mu' - 1$ and $z \geq e' - 1$, and put $c := y + z$ and $V' := ye' + z\mu' - C'$. For $0 \leq t \leq x + y + z$, the subset sums using exactly t elements — *level t* — form $ta + \mathcal{O}_t$, where

$$\mathcal{O}_t \;=\; \{\, je + k\mu \;:\; j \leq y,\ k \leq z,\ \max(0, t - x) \leq j + k \leq t \,\}.$$

Lemma 4.3 (Staircase windows and merges)**.** *In the level decomposition above:*

(a) *if $c \leq t \leq x$ then $\mathcal{O}_t \supseteq g\,[C', V']$;*

(b) (two-frame merge, $g = 1$) *if $x = c + 1$ and $V - C \geq a - 1$ (writing $V = V'$, $C = C'$), the sums cover $[\,ca + C,\ (c+1)a + V\,]$, of length $a + (V - C) + 1$;*

(c) (short two-frame merge, $g = 1$) *if $x = c$, then level $c + 1$ imposes only $j + k \geq 1$, which excludes exactly the offset 0; so $\mathcal{O}_{c+1} \supseteq [\max(C, 1),\ V]$, and if $V \geq a + \max(C, 1) - 1$ the sums cover $[\,ca + C,\ (c+1)a + V\,]$, of length $a + V - C + 1$;*

(d) (g-phase union, $g \geq 2$) *if $x = c + g$ then all levels $t \in [c, c+g]$ satisfy (a); since $\gcd(a, g) = 1$, each residue class modulo g of the target is served by a unique $t_0 \in [c, c+g-1]$, and the sums cover the solid interval $[\,(c+g-1)a + gC',\ ca + gV'\,]$, of length $g(V' - C') - (g-1)a + 1$ (the* base form*: one frame per class, no further hypothesis needed). More generally, if*

$$x \geq c + g \qquad \text{and} \qquad V' - C' \geq a - 1 \quad (\textit{the}\ \text{frame-merge condition}),$$

then within each residue class modulo g the frames $t_0, t_0 + g, t_0 + 2g, \cdots \leq x$ merge, and the sums cover the solid interval $[\,(c + g - 1)a + gC',\ (x - g + 1)a + gV'\,]$ (the extended form*).*

Proof. (a) For $t \in [c, x]$ every (j, k) in the box has $j + k \leq c \leq t$ and $i = t - j - k \in [0, x]$; apply Lemma 4.1 to (e', μ') and scale by g.

(b) Levels c and $c+1$ cover $ta + [C, V]$ each; they merge exactly when $ca + V \geq (c+1)a + C - 1$.

(c) Every nonzero offset of the box keeps all of its representations at level $c+1$, each having $j+k \geq 1$.

(d), *base form.* Every n in the displayed interval satisfies $n \equiv t_0 a \pmod{g}$ for a unique $t_0 \in [c, c+g-1]$, and $u := (n - t_0 a)/g$ satisfies $u \geq C'$ (from the left endpoint, since $t_0 \leq c+g-1$) and $u \leq V'$ (from the right endpoint, since $t_0 \geq c$). By (a), level t_0 realizes $t_0 a + gu = n$.

(d), *extended form.* Fix a residue class modulo g and its unique $t_0 \in [c, c+g-1]$. The levels serving this class within $[c, x]$ are exactly the frames $t_j := t_0 + jg$ for $0 \leq j \leq J$, $J := \lfloor (x - t_0)/g \rfloor \geq 0$ (well defined since $t_0 \leq c+g-1 \leq x-1 < x$, using $x \geq c+g$). By (a), frame t_j covers $t_j a + g[C', V']$, an arithmetic progression of step g inside the class $t_0 a + g\mathbb{Z}$. Two consecutive frames t_j, t_{j+1} leave no gap of the class between them iff $t_{j+1}a + gC' \leq t_j a + gV' + g$, i.e. iff $ga + gC' \leq gV' + g$, i.e. iff $V' - C' \geq a - 1$ — precisely the frame-merge condition, under which the class is covered solidly (in steps of g) on $[\, t_0 a + gC',\ (t_0 + Jg)a + gV' \,]$. When $J = 0$ this holds trivially. Over $t_0 \in [c, c+g-1]$ the left endpoints are all $\leq (c+g-1)a + gC'$; and by maximality of J we have $t_0 + (J+1)g > x$, i.e. $t_0 + Jg > x - g$, so the right endpoints are all $\geq (x-g+1)a + gV'$. Hence every integer of $[\,(c+g-1)a + gC',\ (x-g+1)a + gV'\,]$ lies in the solidly covered range of its own class. □

Remark 4.4 (The merge condition is not removable from the extended form)*.* For $(a, b, M) = (5, 7, 9)$ — so $e = 2$, $\mu = 4$, $g = 2$, $e' = 1$, $\mu' = 2$, $C' = 0$ — take $y = 1$, $z = 0$ (hence $c = 1$, $V' = 1$) and $x = 3 = c + g$. The merge condition fails ($V' - C' = 1 < 4 = a - 1$), and so does the conclusion: the multiset $\{5, 5, 5, 7\}$ has subset sums $\{0, 5, 7, 10, 12, 15, 17, 22\}$, which miss 11 inside the would-be interval $[10, 12]$. An earlier form of Lemma 4.3(d) omitted this hypothesis and was false; the counterexample is pinned as a decided guard in the formalization.

Lemma 4.5 (λ-lift)**.** *Suppose the box (Y, Z) modulo a (Lemma 4.2 with $\nu = a$, $(g_1, g_2) = (b, M)$) covers all residues with budget $\mathcal{B} \leq M - 1$ for the triple (a, b, M), and $Y + Z \leq a - 1$. Then the same box has budget $\leq M' - 1$ for the triple $(a,\ b + a,\ M + a)$, where $M' := M + a$.*

Proof. Residues of $jb + kM$ modulo a are unchanged, so the same representative pairs serve. Each representative height grows by $(j+k)a \leq (Y+Z)a$, so $S' \leq S + (Y+Z)a$ and

$$\left\lceil \frac{M' - 1 + S'}{a} \right\rceil \ \leq\ \left\lceil \frac{M - 1 + S}{a} \right\rceil + Y + Z + 1.$$

Thus the budget grows by at most $Y + Z + 1 \leq a$, while the target $M - 1$ grows by exactly a. □

4.2. **Reduction to the hard core.**

Lemma 4.6 (Graham reduction)**.** *Lemma 1.4 for all alphabets follows from Lemma 1.4 for* irreducible *ones: those with $|G| = 3$, or with $|G| \geq 4$ and G minimal (no proper subset of size ≥ 3 has gcd 1).*

Proof. We use Graham's growth lemma [Gr64]: if the subset sums of a multiset contain a run of ℓ consecutive integers and $g \in G$ satisfies $g \leq \ell$, then adjoining one copy of g extends the run to length $\ell + g$.

Induct on $|G|$. If $|G| \geq 4$ is not irreducible, some $e \in G$ is redundant, i.e. $\gcd(G \setminus \{e\}) = 1$ and $|G \setminus \{e\}| \geq 3$. If $e \neq M$, any witness for $G \setminus \{e\}$ is a witness for G. If $e = M$, let $M' := \max(G \setminus \{M\})$; by induction a run of length $\geq M'$ costs $\leq M' - 1$, and Graham-filling with copies of $a = \min G$ costs $\leq \lceil (M - M')/a \rceil \leq M - M'$, for a total of $\leq M - 1$. □

Lemma 4.7 (Small coprime pair)**.** *If some coprime pair $\alpha, \gamma \in G$ has $\alpha + \gamma \leq M - 1$, then $m(G) \leq M - 1$.*

Proof. Take $x = M - \alpha$ copies of α and $\alpha - 1$ copies of γ. By Lemma 4.2 (modulus α, representatives $j\gamma$) the sums cover $[(\alpha-1)\gamma,\ x\alpha]$, of length $x\alpha - (\alpha-1)\gamma + 1 \geq (M-\alpha)\alpha - (\alpha - 1)(M-\alpha-1) + 1 = M$. The budget is $(M-\alpha) + (\alpha-1) = M-1$. $\square$

Lemma 4.8 (Pair at the boundary)**.** *A coprime pair $\alpha < \beta$ in G with $\alpha + \beta \in \{M, M+1\}$ gives $m(G) \leq M - 1$.*

Proof. For an integer $t \geq 0$, take $x = \beta - 1$ copies of α and $y = \alpha - 1 + t$ copies of β. The hypotheses of Lemma 4.1 hold, so the subset sums contain the interval $[C,\ \alpha x + \beta y - C]$ with $C = (\alpha-1)(\beta-1)$, of length $\alpha x + \beta y - 2C + 1 = \alpha + \beta - 1 + t\beta$, at budget $x + y = \alpha + \beta - 2 + t$. The two boundary values need *different* t:

- If $\alpha + \beta = M + 1$, take $t = 0$: run length $\alpha + \beta - 1 = M$, budget $\alpha + \beta - 2 = M - 1$.
- If $\alpha + \beta = M$, take $t = 1$: run length $\alpha + \beta - 1 + \beta = M - 1 + \beta \geq M$ (as $\beta \geq 2$), budget $\alpha + \beta - 2 + 1 = M - 1$. Here $t = 0$ would give a run of length only $M - 1$; one further copy of β is needed, and the budget absorbs it exactly.

$\square$

Example 4.9. For $(\alpha, \beta, M) = (2, 3, 5)$ (the case $\alpha + \beta = M$), $t = 0$ gives the multiset $\{2, 2, 3\}$ with subset sums $\{0, 2, 3, 4, 5, 7\}$: longest run $[2, 5]$, of length $4 < 5$ — insufficient, exactly as the boundary bookkeeping predicts. With $t = 1$ the multiset is $\{2, 2, 3, 3\}$, of budget $4 = M - 1$, with subset sums $\{0, 2, 3, 4, 5, 6, 7, 8, 10\} \supseteq [2, 8]$: a run of $7 \geq 5$ consecutive integers.

Lemma 4.10 (Non-coprime pair)**.** *If $|G| = 3$ and $a, b \in G \setminus \{M\}$ have $d := \gcd(a, b) \geq 2$ — in which case $\gcd(d, M) = 1$ automatically — then $m(G) \leq M - 1$.*

Proof. Write $a = d\alpha$, $b = d\beta$ with $\gcd(\alpha, \beta) = 1$ and $1 \leq \alpha < \beta$; note $\gcd(d, M) = 1$, since any common factor would divide $\gcd(a, b, M) = 1$.

The multiset and its budget. Use

$$x = \beta - 1 \text{ copies of } a, \qquad y = M - d - \beta + 1 \text{ copies of } b, \qquad d - 1 \text{ copies of } M.$$

Its size is $x + y + (d-1) = (\beta - 1) + (M - d - \beta + 1) + (d - 1) = M - 1$, meeting the budget exactly.

The two-generator grid. As $a = d\alpha$ and $b = d\beta$, the sub-sums using only a's and b's are d times the (α, β)-grid. Lemma 4.1 needs $x \geq \beta - 1$ (here an equality) and $y \geq \alpha - 1$; the latter holds because $M > b = d\beta$ gives $M \geq d\beta + 1$ and so $y \geq (d-1)(\beta-1) + 1 \geq \beta > \alpha - 1$. Writing $C := (\alpha - 1)(\beta - 1)$, that lemma scaled by d yields an arithmetic progression of step d,

$$d \cdot [\,C,\ \alpha x + \beta y - C\,], \qquad \text{of } L := \alpha x + \beta y - 2C + 1 \text{ terms.}$$

The grid is long enough: $L \geq M + 1$. L increases with M (through y), so it suffices to check the smallest admissible $M = d\beta + 1$, where $L \geq M + 1$ reduces to

$$d\beta(\beta - 2) - \beta^2 - \alpha\beta + \alpha + 4\beta - 3 \ \geq\ 0,$$

and for $d \geq 2$ this follows from $\beta(\beta - \alpha) + \alpha - 3 \geq 0$, true for all $1 \leq \alpha < \beta$.

Bridging the d residues. The $d - 1$ copies of M contribute kM for $0 \leq k \leq d - 1$; since $\gcd(d, M) = 1$ these run through all d residues modulo d, so adding them to the step-d grid fills the gaps between consecutive grid points. The result is the solid block $[\,dC + (d-1)M,\ dC + (L-1)d\,]$ of consecutive integers, of length $\geq M$: the requirement $(L-1)d \geq dM - 1$ is, reducing both sides modulo d, exactly $L \geq M + 1$, just shown. $\square$

Lemma 4.11 (Structure of minimal alphabets)**.** *Let $G = \{g_1, \ldots, g_m\}$ be minimal with $m \geq 4$, and put $d_i := \gcd(G \setminus \{g_i\})$. Then each $d_i \geq 2$; the d_i are pairwise coprime; and $d_j \mid g_i$ for every $j \neq i$. Hence every g_i is divisible by $\prod_{j \neq i} d_j$, a product of $m - 1 \geq 3$ pairwise coprime integers*

≥ 2, *so* $g_i \geq 2 \cdot 3 \cdot 5 = 30$ *when* $m = 4$, *and* $g_i \geq 210$ *when* $m \geq 5$. *Also every pair of elements of* G *has gcd* ≥ 2.

Proof. Each $d_i \geq 2$ by minimality. For $i \neq j$, $\gcd(d_i, d_j)$ divides every element of $(G \setminus \{g_i\}) \cup (G \setminus \{g_j\}) = G$, hence divides $\gcd(G) = 1$; so the d_i are pairwise coprime. That $d_j \mid g_i$ for $j \neq i$ is immediate from the definition. Finally a coprime pair would lie in a gcd-1 proper triple, contradicting minimality. ☐

There are exactly 6 minimal 4-alphabets with $M \leq 110$ and 12 with $M \leq 130$, the smallest being $\{30, 42, 70, 105\}$ with $M = 105$; no minimal alphabet of any size has $M < 105$, and $|G| \geq 5$ is impossible below $M = 210$. (This enumeration is machine-checked two independent ways; see Appendix A.)

Lemma 4.12 (Minimal alphabets, $|G| \geq 4$). *If* G *is minimal with* $|G| \geq 4$, *then* $m(G) \leq M - 1$.

Proof. Reduce to a smaller alphabet. Remove the smallest element a. Then $G' := G \setminus \{a\}$ has $\gcd(G') =: d \geq 2$ by minimality, still contains M, and $H := G'/d$ is an alphabet with $\gcd(H) = 1$, $\max H = M/d < M$ and $|H| \geq 3$.

Induction, then Graham-fill. By strong induction on M, a run of $\max H$ consecutive integers is covered within H at cost $\leq M/d - 1$. Put $a' := \min H$ and extend this run by Graham-filling with copies of a' up to length $\ell := \lceil (M - 1 + (d-1)a)/d \rceil + 1$; the extra copies cost $\leq \lceil (\ell - M/d)/a' \rceil$, so a run of ℓ consecutive integers in H costs $\leq (M/d - 1) + \lceil (\ell - M/d)/a' \rceil$.

Scale and bridge. Multiplying by d turns this run into an arithmetic progression of step d; the d residues modulo d are bridged by $k \leq d-1$ copies of a (legitimate since $\gcd(a, d) = 1$, because $\gcd(G) = 1$), which fills the gaps and covers a solid interval of length $\geq M$, at total budget

$$T \;\leq\; \Big(\frac{M}{d} - 1\Big) + \Big\lceil \frac{\ell - M/d}{a'} \Big\rceil + (d-1).$$

Bound the middle term. From $a < \min G' = da'$, i.e. $a \leq da' - 1$, we get $\ell - M/d \leq (d-1)a/d + 2 < (d-1)a' + 2$, hence $\lceil (\ell - M/d)/a' \rceil \leq d + 1$ and $T \leq M/d + 2d - 1$.

Close. Then

$$T \leq M - 1 \iff M\Big(1 - \tfrac{1}{d}\Big) \geq 2d \iff M \geq \frac{2d^2}{d-1}.$$

For $d \geq 3$, $2d^2/(d-1) \leq 3d \leq M$, since G' contains at least three distinct multiples of d. For $d = 2$ the requirement is $M \geq 8$, which holds since every element is ≥ 30 (Lemma 4.11). The bound uses a' only through $a \leq da' - 1$, so the case $a' = 1$ is subsumed. ☐

4.3. **The hard core.** By Lemmas 4.6–4.12, Lemma 1.4 reduces to triples $G = \{a, b, M\}$ with $\gcd(a, b) = 1$ and — applying Lemmas 4.7 and 4.8 to the pair (a, b) — with $a + b - M \geq 2$. In the standing notation of §4.1, the *hard core* is thus the set of triples with

$$e := b - a \geq 1, \quad h := M - b \in [1, a-2], \quad \mu := e + h = M - a, \quad \gcd(a, e) = 1, \quad a \geq 3,$$

with $g = \gcd(e, h) = \gcd(e, \mu)$ coprime to each of a, b, M. Additionally set $\bar{e} := e \bmod a \in [1, a-1]$, $\bar{\mu} := M \bmod a$, and — when $\bar{\mu} \neq 0$ — $\eta := \bar{\mu}\, \bar{e}^{-1} \bmod a$. As a mnemonic: *bars* are mod-a reductions, *primes* are divisions by g.

These symbols recur densely for the rest of the section, so we collect them.

Symbol	Meaning	Symbol	Meaning
$a < b < M$	the hard-core triple, $\gcd(a,b) = 1$	g	$\gcd(e,h) = \gcd(e,\mu)$
e	$b - a \geq 1$	e', μ'	e/g, μ/g (coprime)
h	$M - b \in [1, a-2]$	C'	$(e'-1)(\mu'-1)$
μ	$e + h = M - a$	V'	$ye' + z\mu' - C'$
t	$\min(\eta,\ a-\eta)$ (Case E)	(x,y,z)	copies of a, b, M
$\bar{e}$	$e \bmod a \in [1, a-1]$	c	$y + z$
$\bar{\mu}$	$M \bmod a$	$\mathcal{B}$	budget $= x + y + z$
η	$\bar{\mu}\,\bar{e}^{-1} \bmod a$	λ	$\lfloor e/a \rfloor$ (the lift index)

Note that $\eta = 1$ is impossible ($\eta = 1 \iff a \mid h$, but $1 \leq h \leq a-2$), and that

$$\eta = 0 \iff a \mid M, \qquad \eta = a-1 \iff a \mid b + M,$$

these two lines being disjoint (both would force $a \mid b$). The target budget is $M - 1$ throughout.

Each hard-core triple is routed to exactly one of six branches — the ordered if–else tree of Table 1, evaluated top to bottom, treated in §§4.4–4.9. The branch *conditions* can overlap — e.g. $(3,4,5)$ satisfies both P and L — but the ordered routing assigns every triple to exactly one branch; exhaustiveness is checked in §4.10. Each branch yields a covering multiset of size $\leq M-1$. In the table, "symbolic" means the branch is closed by a closed-form construction valid for all parameters in its range, and "finite" means it additionally consumes a certificate table.

Branch	Condition	Construction	Budget mechanism	Closure
D *(divisible)*	$a \mid M$	Lemma 4.13	$(a-1)$ copies of b span $\mathbb{Z}_a$; a, M pad at "half price"	symbolic
P *(paired)*	$a \mid b + M$	Lemma 4.14	(b,M) pairs are residue-neutral movers worth ra	symbolic
L *(linear)*	$e = h$	Lemma 4.17	staircase, extended form	symbolic
E *(η-box)*	$a, \mu \geq 12$	Lemma 4.18	the η-box $(t-1, Z)$	symbolic
T *(thin μ)*	$\mu \leq 11$	Lemma 4.20, Prop. 4.22	staircase + linear tail; 172 exceptions	finite (Table A)
B *(bounded a)*	$a \leq 11$, $\mu \geq 12$	Table B + Lemma 4.5	finite bases, lifted in λ	finite (Table B)

TABLE 1. The six-branch case tree for the hard core: routing is the ordered if–else evaluation, top to bottom. "Symbolic" branches close by a closed-form construction; "finite" branches also consume a certificate table.

Four of the six branches are closed symbolically; only T and B consume finite data, and between them they account for the whole of the finite layer (Proposition A.1).

4.4. **Case D:** $a \mid M$**.**

Lemma 4.13. *If $a \mid M$, write $M = qa$ with $q \geq 2$ and put $x_{\mathrm{eff}} := \lceil (M - 1 + (a-1)b)/a \rceil$. Then the multiset with $a-1$ copies of b, $q-1$ copies of a, and $z := \lceil (x_{\mathrm{eff}} - (q-1))/q \rceil$ copies of M has budget $\leq M-1$, and its subset sums contain M consecutive integers.*

Proof. Coverage. The $a-1$ copies of b realize $\{jb : 0 \leq j \leq a-1\}$, a complete residue system modulo a (as $\gcd(a,b) = 1$), with top $S = (a-1)b$. The padding realizes $\{ia + kM : 0 \leq i \leq q-1,\ 0 \leq k \leq z\} = a\{i + qk\} \supseteq a\,[0,\ (q-1) + qz]$, because the blocks $[qk,\ qk + (q-1)]$ abut, each of length q; and $(q-1) + qz \geq x_{\mathrm{eff}}$ by the choice of z. By Lemma 4.2 the subset sums cover $[S,\ S + M - 1]$.

Budget. Since $b \leq M-1$,

$$\frac{M-1+(a-1)b}{a} \;\leq\; \frac{M-1+(a-1)(M-1)}{a} \;=\; M-1,$$

so $x_{\text{eff}} \leq M-1 = qa-1$, whence

$$z \;=\; \left\lceil \frac{x_{\text{eff}}-(q-1)}{q} \right\rceil \;\leq\; \left\lceil \frac{qa-1-(q-1)}{q} \right\rceil \;=\; \left\lceil \frac{q(a-1)}{q} \right\rceil \;=\; a-1.$$

Therefore

$$(a-1)+(q-1)+z \;\leq\; (a-1)+(q-1)+(a-1) \;=\; 2a+q-3 \;\leq\; qa-1 \;=\; M-1,$$

the last inequality being exactly $(a-1)(q-2) \geq 0$, valid for all $q \geq 2$ and $a \geq 3$. A single construction therefore suffices for the whole case; no split on q is needed. □

4.5. **Case P: $a \mid b+M$.** Here $b+M = ra$ for an integer $r \geq 3$, and copies of b and M act as opposite residue steps modulo a: a (b, M)-pair is a residue-neutral mover worth r grid units of a at a cost of 2.

Lemma 4.14 (Uniform pair-frame)**.** *Let (a, b, M) be a hard-core triple with $b+M = ra$, $r \geq 3$. Then the multiset*

$$(x, y, z) \;=\; \Big(r-1,\; \left\lceil \frac{M-r}{2} \right\rceil,\; \left\lfloor \frac{M-r}{2} \right\rfloor\Big)$$

of copies of (a, b, M), whose budget is exactly $(r-1)+\lceil (M-r)/2 \rceil + \lfloor (M-r)/2 \rfloor = M-1$, has subset sums containing M consecutive integers.

No restriction on r or a is imposed: a single closed-form multiset serves the whole line.

Proof. Any selection from the multiset is

$$v \;=\; ia+jb+kM \;=\; ia+wb+rak, \qquad w := j-k$$

(using $M = ra-b$), subject to $i \leq x = r-1$, $k \geq \max(0, -w)$, $k \leq z$ and $w+k \leq y$. So a (b, M) pair is a residue-neutral mover worth ra, and w is the number of *unpaired* copies of b (if $w > 0$) or of M (if $w < 0$).

Set

$$p := \left\lfloor \frac{a}{2} \right\rfloor, \qquad q := \left\lfloor \frac{a-1}{2} \right\rfloor, \qquad \text{so} \qquad p+q = a-1,$$

and fix the run start $\text{low} := \max(p\,b,\; q\,M)$. We show that every $n \in [\text{low},\; \text{low}+M-1]$ is realized.

Preliminaries. We record three consequences of the hypotheses, each valid for all $a \geq 3$ and $r \geq 3$.

- (P1) $M \geq a+r-1$. From $b < M$ and $b+M = ra$ we get $ra \leq 2M-1$. Since $a \geq 3$ and $r \geq 3$ give $a(r-2) \geq 3(r-2) \geq 2r-3$, it follows that $2M \geq ra+1 = a(r-2)+2a+1 \geq 2a+2r-2$.
- (P2) $M \leq a(r-1)-1$. From $b > a$ we get $M = ra-b < ra-a = a(r-1)$.
- (P3) *$y \geq p$ and $z \geq q$.* By (P1), $M-r \geq a-1$, so $z = \lfloor (M-r)/2 \rfloor \geq \lfloor (a-1)/2 \rfloor = q$ and $y = \lceil (M-r)/2 \rceil \geq \lceil (a-1)/2 \rceil = \lfloor a/2 \rfloor = p$.

Residues, and where the four inequalities come from. Since $\gcd(a, b) = 1$, for each target $n \in [\text{low},\; \text{low}+M-1]$ there is a unique $w \in [0, a-1]$ with $wb \equiv n \pmod{a}$. We split on the size of w, and in each branch we exhibit the multiset explicitly and read off exactly what must be checked.

Branch 1: $w \leq p$ (spend w unpaired copies of b). Here $wb \leq pb \leq \text{low} \leq n$, so $A := (n-wb)/a$ is a nonnegative integer. Write $A = i + r\kappa$ with $i := A \bmod r$, so $i \leq r-1 = x$ and $\kappa := \lfloor A/r \rfloor$.

Take the sub-multiset consisting of i copies of a, $w+\kappa$ copies of b, and κ copies of M. Its value is

$$ia+(w+\kappa)b+\kappa M \;=\; ia+wb+\kappa(b+M) \;=\; ia+wb+\kappa ra \;=\; a(i+r\kappa)+wb \;=\; aA+wb \;=\; n,$$

as required. It is available in the multiset (x,y,z) precisely when

$$\kappa \le z \qquad \text{and} \qquad w+\kappa \le y.$$

For the first: $aA = n-wb \le n \le \mathrm{low}+M-1$, and $\kappa \le z$ follows from $A \le rz+x$, so it suffices that $\mathrm{low}+M-1 \le a(rz+x)$. For the second: using $a\,r\,w = w(b+M)$,

$$a\,(A+rw) \;=\; (n-wb)+w(b+M) \;=\; n+wM \;\le\; (\mathrm{low}+M-1)+pM,$$

and $w+\kappa \le y$ follows from $A+rw \le ry+x$; so it suffices that

$$\mathrm{low}+M-1+p\,M \;\le\; a\,(ry+x). \tag{C$_y$}$$

Branch 2: $w>p$ (spend $v:=a-w\in[1,q]$ unpaired copies of M). Since $M\equiv -b \pmod a$, we have $vM\equiv -vb\equiv wb\equiv n \pmod a$, and $vM\le qM\le \mathrm{low}\le n$, so $A' := (n-vM)/a$ is a nonnegative integer. Writing $A' = i'+r\kappa'$ as before, the sub-multiset of i' copies of a, κ' copies of b and $v+\kappa'$ copies of M has value n, and is available precisely when $\kappa'\le y$ and $v+\kappa'\le z$. The first follows from $\mathrm{low}+M-1\le a(ry+x)$, which is weaker than (C$_y$). For the second, $a(A'+rv) = (n-vM)+v(b+M) = n+vb \le (\mathrm{low}+M-1)+qb$, so it suffices that

$$\mathrm{low}+M-1+q\,b \;\le\; a\,(rz+x). \tag{C$_z$}$$

(This also implies the condition $\mathrm{low}+M-1\le a(rz+x)$ needed in Branch 1.)

The four inequalities. So the entire coverage claim reduces to the two conditions (C$_y$) and (C$_z$). Each contains $\mathrm{low}=\max(pb,\ qM)$, so each splits into two cases according to which term attains the maximum; using $p+q=a-1$ and $b+M=ra$, the four cases are exactly

$$\begin{array}{lll} \text{(E1)} & (a-1)b+M-1 \;\le\; a\,(rz+x) & [\,(\mathrm{C}_z) \text{ with } \mathrm{low}=pb\,],\\ \text{(E2)} & q(b+M)+M-1 \;\le\; a\,(rz+x) & [\,(\mathrm{C}_z) \text{ with } \mathrm{low}=qM\,],\\ \text{(F1)} & p(b+M)+M-1 \;\le\; a\,(ry+x) & [\,(\mathrm{C}_y) \text{ with } \mathrm{low}=pb\,],\\ \text{(F2)} & aM-1 \;\le\; a\,(ry+x) & [\,(\mathrm{C}_y) \text{ with } \mathrm{low}=qM\,]. \end{array} \tag{19}$$

(For (E1): $pb+qb+M-1=(a-1)b+M-1$. For (F2): $qM+pM+M-1=(a-1)M+M-1=aM-1$.) Checking these four extremes suffices for *all* w and all n in the target interval, because the only place w entered was through the bounds $w\le p$ and $v\le q$, which are exactly the extremes used. We now verify all four.

(F2). It suffices that $M\le ry+r-1$. From $y\ge (M-r)/2$ and $r\ge 3$,

$$ry+r-1 \;\ge\; \tfrac{3}{2}(M-r)+r-1 \;=\; \tfrac{1}{2}(3M-r-2) \;\ge\; M \iff M\ge r+2,$$

and $M\ge a+r-1\ge r+2$ by (P1) and $a\ge 3$.

(F1). Here $p(b+M)=p\,ra$, and $y\ge p$ by (P3) gives $ary\ge p\,ra$. So (F1) reduces to $M-1\le a(r-1)$, which is (P2).

(E2). Identically, $q(b+M)=q\,ra$, and $z\ge q$ by (P3) gives $arz\ge q\,ra$, so (E2) also reduces to (P2).

(E1). This is the binding inequality. Substituting $b=ra-M$ and using $z\ge (M-r-1)/2$, it suffices that

$$2\big[(a-1)(ra-M)+M-1\big] \;\le\; ar(M-r-1)+2a(r-1).$$

Expanding, the left side is $2ra^2 - 2aM - 2ra + 4M - 2$ and the right side is $arM - ar^2 + ar - 2a$, so the inequality rearranges to

$$0 \leq M\,(ar + 2a - 4) - ar^2 - 2ra^2 + 3ar - 2a + 2.$$

The coefficient $ar+2a-4$ is positive (as $a, r \geq 3$), so we may insert the lower bound $2M \geq ra+1$ from (P1). Doing so and multiplying through by 2, the right-hand side is at least

$$(ra+1)(ar+2a-4) - 2ar^2 - 4ra^2 + 6ar - 4a + 4 = a\big(ar^2 - 2ar - 2r^2 + 3r - 2\big).$$

Hence (E1) holds as soon as

$$a\,r\,(r-2) \geq 2r^2 - 3r + 2. \tag{20}$$

For $r \geq 4$: since $a \geq 3$ we have $a\,r(r-2) \geq 3r^2 - 6r$, and $3r^2 - 6r \geq 2r^2 - 3r + 2 \iff r^2 - 3r - 2 \geq 0$, which holds for every $r \geq 4$. For $r = 3$, (20) reads $3a \geq 11$, i.e. $a \geq 4$.

So (20) can fail only for $(a, r) = (3, 3)$. But $a = 3$ together with $b + M = 3a = 9$ and $3 < b < M$ forces $b = 4$ and $M = 5$: the single triple $(3, 4, 5)$, for which $(x, y, z) = (2, 1, 1)$ and (E1) reads $2 \cdot 4 + 5 - 1 = 12 \leq 3(3 \cdot 1 + 2) = 15$, true. (Concretely, the multiset $\{3, 3, 4, 5\}$ has subset sums $\{0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15\} \supseteq [3, 12]$, a run of $10 \geq 5$ consecutive integers, at budget $4 = M - 1$.)

All four inequalities of (19) therefore hold for every hard-core triple on the line $a \mid b + M$, which completes the proof. □

Proposition 4.15 (Exhaustion of Case P)**.** *Every hard-core triple with $a \mid b + M$ satisfies $m(G) \leq M - 1$.*

Proof. Write $b + M = ra$. Since $a < b < M$ we have $b + M > 2a$, so $r \geq 3$, and Lemma 4.14 applies. (Note $r = 2$, i.e. $M = 2a - b$, is impossible, as is $M = 2a$, which would force $a \mid b$.) □

Remark 4.16. Case P therefore requires no subset-sum search and no certificate table: a single closed-form multiset serves every triple on the line, and the only triple not covered by the estimate (20) is $(3, 4, 5)$, which is checked directly. In particular the three triples $(3, 7, 8)$, $(4, 9, 11)$, $(5, 12, 13)$ — which earlier treatments carried as ad-hoc certificates for the $r \geq 5$, $M < 2a + 4$ corner — are ordinary instances of it, with $x = r - 1 = 4$.

The formalization proves this case by a slightly different route; the deviation, and the short auxiliary lemma it uses, are recorded in §6.5.

4.6. Case L: $e = h = g$.

Lemma 4.17. *For $G = \{a,\ a+g,\ a+2g\}$ with $\gcd(a, g) = 1$ and $g \leq a - 2$, the multiset*

$$(x, y, z) = \Big(\Big\lfloor \frac{a-2}{2} \Big\rfloor + 2g,\quad 1,\quad \Big\lceil \frac{a-2}{2} \Big\rceil\Big),$$

of budget $a + 2g - 1 = M - 1$, covers M consecutive integers.

Proof. Here $\mu = 2g$, $(e', \mu') = (1, 2)$, $C' = 0$, $V' = 2z + 1$, $c = z + 1$. The offsets are $je + k\mu = g(j + 2k)$, and with $y = 1$ we get $\{j + 2k : j \leq 1,\ k \leq z\} = [0,\ 2z + 1]$, contiguous. Note $x - c = 2g - 2$ for odd a and $2g - 1$ for even a. Three sub-cases.

a even. Here $x - c = 2g - 1 \geq g$, so $x \geq c + g$; and with $C' = 0$, $V' = 2z + 1$, the frame-merge condition of Lemma 4.3(d) reads $V' - C' = 2z + 1 \geq a - 1$, i.e. $z \geq (a-2)/2$ — exactly our $z = (a-2)/2$, with equality. Both hypotheses of the extended form hold, so the sums cover the solid interval $[\,(c + g - 1)a,\ (x - g + 1)a + g(2z + 1)\,]$, of length $a + g(a - 1) + 1 \geq M = a + 2g$, since $g(a - 1) \geq 2g - 1$ for $a \geq 3$.

a odd, $g \geq 2$. Here $z = (a-1)/2$ and $x = c + 2(g - 1) \geq c + g$ (as $g \geq 2$), and the frame-merge condition holds with room to spare: $V' - C' = 2z + 1 = a \geq a - 1$. The extended form gives the

solid interval $[\,(c+g-1)a,\ (x-g+1)a+g(2z+1)\,]$, of length $g(2z+1)+1 = ga+1 \geq a+2g$, since $(g-1)a \geq 2g-1$ for $g \geq 2$, $a \geq 3$.

a odd, $g=1$. Here $x=c$, so Lemma 4.3(c) applies: level c covers $ca+[0,V]$ with $V = 2z+1 = a$, level $c+1$ covers $(c+1)a+[1,V]$, and the merge condition $V \geq a+\max(C,1)-1 = a$ holds with equality. The run $[\,ca,\ (c+1)a+V\,]$ has length $2a+1 \geq a+2 = M$, at budget $2c = a+1 = M-1$. □

4.7. **Case E: the universal η-box.** Recall $\bar{e} = e \bmod a$, $\bar{\mu} = M \bmod a$ and $\eta = \bar{\mu}\,\bar{e}^{-1} \bmod a$ (bars: mod-a reductions); this case works entirely modulo a.

Lemma 4.18. *Suppose $a \nmid M$ and $a \nmid b+M$ (so $\eta \notin \{0,1,a-1\}$), and suppose $a \geq 12$ and $\mu = M-a \geq 12$. Let $t := \min(\eta,\ a-\eta) \in [2,\ a/2]$, $Z := \lceil (a-t)/t \rceil$ and $\sigma := t+Z$. Then the box $(Y,Z) = (t-1,\ Z)$ covers $\mathbb{Z}_a$ and its budget is $\leq M-1$.*

Proof. Coverage. If $t=\eta$, the box offsets are $\bar{e}(j+tk) \bmod a$ and $\{j+tk : j \leq t-1,\ k \leq Z\} = [0,\ t(Z+1)-1] \supseteq [0,a-1]$, since $t(Z+1) \geq a$. If $t = a-\eta$, the offsets are $\bar{e}(j-tk)$ and $\{j-tk\} = [-tZ,\ t-1]$, an interval of length $t(Z+1) \geq a$. Either way we obtain a complete residue system, multiplication by the unit $\bar{e}$ being a bijection.

Budget. Write $S_0 := (t-1)b+ZM$ for the corner bound on the representative height, so $S \leq S_0$, and put $K := t-1+Z = \sigma-1$. Then

$$\mathcal{B} \ \leq\ \left\lceil \frac{M-1+S_0}{a} \right\rceil + K,$$

so it *suffices* that $M-1+S_0 \leq a(M-1-K)$: this forces $\lceil (M-1+S_0)/a \rceil \leq M-1-K$ and hence $\mathcal{B} \leq M-1$. Using $b \leq M-1$ we have $M-1+S_0 \leq M\sigma - t$, while $a(M-1-K) = a(M-\sigma)$, so the sufficient inequality is in turn implied by $M\sigma + a\sigma \leq aM+t$, and hence by $M(a-\sigma) \geq a\sigma$.

Threshold. We have $\sigma \leq t+(a-1)/t$, whose maximum over $t \in [2,a/2]$ is attained at an endpoint, by convexity: $\sigma \leq \max\left(\frac{a+3}{2},\ \frac{a}{2}+2\right) = \frac{a}{2}+2$. Hence

$$\frac{a\sigma}{a-\sigma} \ \leq\ \frac{a(a+4)}{a-4} \ =\ a+8+\frac{32}{a-4} \ \leq\ a+12 \qquad (a \geq 12),$$

so $M = a+\mu \geq a+12$ suffices. □

Example 4.19 (a Case-E instance). Take $(a,b,M) = (13,22,27)$: here $e=9$, $h=5$, $\mu = 14 \geq 12$, $\bar{e} = 9$, $\bar{\mu} = 27 \bmod 13 = 1$, and $\eta = \bar{\mu}\,\bar{e}^{-1} = 9^{-1} = 3$ in $\mathbb{Z}_{13}$ (as $9 \cdot 3 = 27 \equiv 1$). Then $t = \min(\eta, a-\eta) = 3$, $Z = \lceil 10/3 \rceil = 4$, and the box is $(Y,Z) = (2,4)$: the offsets $\bar{e}(j+3k)$ with $j \leq 2$, $k \leq 4$ have $j+3k$ running over $[0,14] \supseteq [0,12]$, a complete residue system modulo 13. The corner bound is $S_0 = 2b+4M = 152$, so the budget is at most $\lceil (26+152)/13 \rceil + 6 = 14+6 = 20 \leq 26 = M-1$.

4.8. **Case T: $\mu \leq 11$.** Fix (e,h) with $\mu = e+h \leq 11$, $e \neq h$ (else Case L), $a \nmid M$ and $a \nmid b+M$. Since $e \leq 10$ and $h \leq 10$, this case consists of finitely many *lines*, each parameterized by a alone. Recall $e' = e/g$ and $\mu' = \mu/g$ (primes: division by $g = \gcd(e,\mu)$); as $e \neq h$ and $e' < \mu'$, here $\mu' \geq 3$.

Construction T.. In each variant, z is a maximum of two branches with distinct jobs: the constant branch keeps $z \geq e'-1$ (the standing hypothesis of the level decomposition), and the ceiling branch buys run length $\geq M$. Take $y = \mu'-1$ (writing $C := C'$ and $V := V'$ when $g=1$, where $e' = e$ and $\mu' = \mu$, so $C = (e-1)(\mu-1)$), and:

- *variant A* $(g=1)$: $z = \max\left(e-1,\ \lceil (\max(a,\mu)-1+2C-ye)/\mu \rceil\right)$ and $x = c+1$;
- *variant* $g \geq 2$: $z = \max\left(e'-1,\ \lceil (a+\lceil(\mu-1)/g\rceil + 2C' - ye')/\mu' \rceil\right)$ and $x = c+g$,

with $c = y+z$ throughout, exactly one variant applying to each line.

Lemma 4.20. *On every Case-T line, Construction T covers M consecutive integers, at budget $2c+v$ with $v \in \{1, g\}$. Both variants are* merge-robust *— they never invoke the short merge, Lemma 4.3(c): variant A uses the two-frame merge of Lemma 4.3(b), whose merge condition its choice of z enforces, and the $g \geq 2$ variant uses the base form of Lemma 4.3(d), which needs no merge hypothesis.*

Proof. Both variants instantiate Lemma 4.3 with the multiset (x, y, z) and $y = \mu' - 1$, whose budget is $x + y + z$; we verify the hypotheses and the run length. Throughout, $V' - C' = ye' + z\mu' - 2C'$, and the two branches of the maximum defining z play distinct roles: the constant branch keeps $z \geq e' - 1$, the standing hypothesis of Lemma 4.3, while the ceiling branch supplies the run length.

Variant A ($g = 1$; here $e' = e$, $\mu' = \mu$, $C = C'$). The ceiling branch gives $z\mu \geq \max(a, \mu) - 1 + 2C - ye$, hence

$$V - C \;=\; ye + z\mu - 2C \;\geq\; \max(a, \mu) - 1 \;\geq\; a - 1,$$

which is exactly the two-frame merge condition of Lemma 4.3(b). With $x = c + 1$, that part covers $[\,ca + C,\ (c+1)a + V\,]$, an interval of length $a + (V - C) + 1 \geq a + \max(a, \mu) \geq a + \mu = M$; so the subset sums contain M consecutive integers. The budget is $x + y + z = 2c + 1$.

Variant $g \geq 2$. The ceiling branch gives $z\mu' \geq a + \lceil(\mu - 1)/g\rceil + 2C' - ye'$, hence

$$V' - C' \;=\; ye' + z\mu' - 2C' \;\geq\; a + \left\lceil \tfrac{\mu-1}{g} \right\rceil \;\geq\; a + \tfrac{\mu-1}{g}.$$

With $x = c + g$, the base form of Lemma 4.3(d) — which needs *no* merge hypothesis — covers a solid interval of length

$$g(V' - C') - (g-1)a + 1 \;\geq\; g\Big(a + \tfrac{\mu-1}{g}\Big) - (g-1)a + 1 \;=\; a + \mu \;=\; M.$$

The budget is $x + y + z = 2c + g$.

Since $g = \gcd(e, \mu) \geq 1$, exactly one variant applies to each line, and both invoke only parts (b) and (d) of Lemma 4.3, never the short merge (c). The coverage hypotheses above hold for every a; the only quantity that can violate the certificate is the budget $2c + v$ ($v \in \{1, g\}$), through its dependence on z. ☐

Only the budget inequality $2c + v \leq M - 1$ can fail, and it fails only finitely often.

Example 4.21 (a Case-T line). Take $(e, h) = (2, 1)$, so $\mu = 3$, $g = \gcd(2, 1) = 1$, $(e', \mu') = (2, 3)$ and $C' = (e' - 1)(\mu' - 1) = 2$. This is a single line, parameterized by a: its members are the hard-core triples $(a,\ a+2,\ a+3)$. Variant A takes $y = \mu' - 1 = 2$ and

$$z \;=\; \max\Big(e - 1,\ \Big\lceil \tfrac{\max(a,\mu) - 1 + 2C' - ye}{\mu} \Big\rceil\Big) \;=\; \max\Big(1,\ \Big\lceil \tfrac{a-1}{3} \Big\rceil\Big),$$

with $c = y + z$ and $x = c + 1$, for a budget of $2c + 1$ against the target $M - 1 = a + 2$. Selected values along the line (the table is a sample, not exhaustive):

a	z	c	budget $2c+1$	target $a+2$	margin
7	2	4	9	9	0 (tight)
13	4	6	13	15	2
19	6	8	17	21	4
25	8	10	21	27	6
31	10	12	25	33	8

The budget grows with slope $2/\mu' = 2/3$ while the target grows with slope 1, so the margin increases at rate $1/3$ and, once positive, stays positive. On this particular line the budget in fact never exceeds the target — it is merely tight at $a = 7$ — so the line contributes *no* rows to

Table A. Other lines are less lucky: the line $(e,h)=(5,1)$ contributes the Table-A rows $(4,9,10)$, $(7,12,13)$ and $(8,13,14)$, among others, where the budget does exceed $M-1$ and an explicit box certificate is needed. Proposition 4.22 is exactly the statement that, over all lines and all $a \le 3000$, this happens for precisely 172 triples, and never for $a > 3000$.

Proposition 4.22 (T-tail)**.** *On every Case-T line, the budget of the applicable variant of Construction T exceeds $M-1$ for precisely 172 triples, all with $a \le 29$ — these are Table A of Appendix B, each carrying a verified modulo-a box certificate with $Y+Z \le a-1$ and budget $\le M-1$ — and for no triple with $a > 3000$. Hence the failure set of the Case-T construction is exactly Table A.*

Proof (computer-assisted for $a \le 3000$). The finite range. The applicable variant is determined by the line: variant A when $g=1$, the base form when $g \ge 2$; the scan evaluates that variant's budget and no other. For every line and every $a \le 3000$, this budget was computed exactly by the finite enumeration of Appendix A (Proposition A.1(iii)); it exceeds $M-1$ for precisely the 172 triples of Table A, all with $a \le 29$. The failing triples are re-verified row-by-row by `gen-tables.py`, decided by the Lean kernel, and each carries its box certificate per Proposition A.1(i)–(ii).

The tail $a > 3000$. For each line, the applicable variant's budget is bounded above by the linear function

$$\beta(a) \ := \ 2\Big[(\mu'-1) + \frac{a+\mu+2C'-(\mu'-1)e'}{\mu'} + 1\Big] + g,$$

of slope $2/\mu' \le 2/3$ in a (using $\lceil u \rceil \le u+1$ and $\mu' \ge 3$), while the target $\tau(a) = a+\mu-1$ has slope 1.

Why β bounds the budget. The z-formulas of Construction T are maxima, and β bounds their ceiling branch; that branch is the *active* one throughout $a \ge 3000$. Indeed, on every line the ceiling's argument is at least $2899/11 > 260$ — the numerator is at least $3000-1-ye \ge 2899$, since $y \le \mu'-1 \le 10$ and $e' \le e \le 10$ give $ye \le 100$, and the denominator is $\mu' \le 11$ — whereas the competing branch, $e-1$ respectively $e'-1$, is at most 9. Being increasing in a against a constant, the ceiling branch stays active for all $a \ge 3000$, so there the budget equals its ceiling value and β bounds it above. The margin $\tau - \beta$ is therefore nondecreasing in a, at rate at least $1/3$.

It remains to check that the margin is positive at $a = 3000$. We compare the target against β, the *proved upper bound* on the applicable variant's budget, and not against the exact budget; no separate reconciliation of exact budgets is therefore required. Evaluating $\beta(3000)$ on each of the finitely many lines and minimizing gives

$$\min_{\text{lines}} \big(\tau(3000) - \beta(3000)\big) \ = \ 995 \ > \ 0,$$

as reported by the verification harness of Appendix A. Since $\tau - \beta$ is nondecreasing, it stays positive for every $a > 3000$. Hence there are no budget failures with $a > 3000$. (The cutoff 3000 is convenience, not necessity: any threshold at which every line's margin is positive would serve. It sits comfortably past the last actual failure at $a = 29$, and is small enough for the kernel-decided sweep.) □

4.9. **Case B: finite bases and the λ-lift.** For $a \le 11$ the classes $(a, \bar{e}, h)$ with $\bar{e} \in [1, a-1]$ and $h \in [1, a-2]$ are finite in number, and the members of a class differ only in $\lambda = \lfloor e/a \rfloor$. For every class reaching this case, the member of smallest M is certified by an explicit modulo-a box with $Y+Z \le a-1$ and budget $\le M-1$ — Table B of Appendix B, comprising 178 classes — and by Lemma 4.5 the same box certifies every larger member of the class.

Which branch a triple belongs to is class-invariant for D and P, since $M \bmod a$ and $(b+M) \bmod a$ depend only on $(\bar{e}, h)$; and an $e = h$ member can only be the $\lambda = 0$ member. Table B lists bases accordingly, taking the first branch-B member of each class.

Example 4.23 (a Case-B class and its lift). Take the class $[a; \bar{e}, h] = [5; 1, 1]$. Its base, listed in Table B, is $(a, b, M) = (5, 16, 17)$ with box $(x, Y, Z) = (10, 1, 2)$: budget $10 + 1 + 2 = 13 \leq M - 1 = 16$. The members of the class are obtained by increasing $\lambda = \lfloor e/a \rfloor$, i.e. by $(a, b, M) \mapsto (a,\ b+5,\ M+5)$, giving $(5, 21, 22)$, $(5, 26, 27)$, and so on — all with the same $b \bmod 5$ and the same $h = M - b = 1$.

By Lemma 4.5 the *same* box $(Y, Z) = (1, 2)$ certifies every one of them: the residues of $jb+kM$ modulo 5 are unchanged, so the same representative pairs serve; each representative height grows by at most $(Y + Z)\,a = 3 \cdot 5 = 15$, so the budget grows by at most $Y + Z + 1 = 4 \leq a = 5$, while the target $M - 1$ grows by exactly $a = 5$. The margin therefore never shrinks, and one finite check at the base settles the whole infinite class. This is why Table B lists 178 *classes* rather than infinitely many triples.

Lemma 4.24 (Descent)**.** *Fix a class $(a,\ \bar{e},\ h)$. Membership of Cases D and P is the same for every member of the class; a member with $e = h$ is necessarily its $\lambda = 0$ (smallest) member; and iterating the descent $b \mapsto b - a$, $M \mapsto M - a$ from any member lands on the class base $b_0 \leq b$ with $b \equiv b_0 \pmod a$, from which Lemma 4.5 transports the base's box upward to the member. Consequently, basing each class reaching Case B at its first branch-B member certifies the entire class.*

Proof. The class-level branch conditions — $M \bmod a$, $(b + M) \bmod a$, and the box's residue coverage — are invariant under the descent $b \mapsto b-a$, $M \mapsto M-a$: here $\gcd(a, b-a) = \gcd(a, b)$, the modulo-$a$ quantities are unchanged, and $h = M - b$ is fixed. So D- and P-membership is class-invariant. The condition $e = h$ is *not* invariant: $e = b-a$ drops by a at each step while h is fixed, so a member with $e = h$ can only be the $\lambda = 0$ member — which then belongs to Case L, its higher members $(e > h)$ to Case B. Finally, the descent terminates at the base ($\lambda = 0$ or the first branch-B member), and Lemma 4.5 lifts its certificate back up, one step of a at a time. □

Concretely, the enumeration for $a \leq 11$ has 178 distinct classes, of which exactly six — the diagonal $\bar{e} = h$ classes $(9, 25, 32)$, $(10, 27, 34)$, $(11, 28, 34)$, $(11, 29, 36)$, $(11, 30, 38)$, $(11, 31, 40)$ — must be based at a $\lambda \geq 1$ member, because their $\lambda = 0$ member has $e = h$ and so belongs to Case L.

4.10. **Proof of Lemma 1.4.**

Proof. We argue by strong induction on M: in proving the statement for a given M we may assume it for every alphabet of smaller maximum. This is what Lemma 4.12 consumes, passing to the rescaled alphabet $H = G'/d$ of maximum $M/d < M$ (in the notation of that lemma). No other branch appeals to the induction hypothesis, and no hard-core branch re-enters Lemma 4.12 at a larger M, so the recursion is well founded.

By Lemmas 4.6–4.12 it suffices to treat the hard core of §4.3. Given a hard-core triple (a, b, M): if $a \mid M$, apply Lemma 4.13; else if $a \mid b + M$, apply Lemma 4.14 (Proposition 4.15); else if $e = h$, apply Lemma 4.17; else if $a \geq 12$ and $\mu \geq 12$, apply Lemma 4.18; else if $\mu \leq 11$, apply Lemma 4.20 and Proposition 4.22 together with Table A; else $a \leq 11$ and $\mu \geq 12$, and we apply Table B with Lemma 4.5.

Exhaustiveness. Cases D and P are disjoint lines (§4.3). A triple with $e = h$ and $a \mid b + M$ would need $a \mid 3e$, possible only for $a = 3$, $e = 1$, i.e. $(3, 4, 5)$, where both Lemma 4.14 (witness $(2, 1, 1)$) and Lemma 4.17 apply. Every remaining triple has $\eta \notin \{0, 1, a-1\}$ and falls into exactly one of E, T, B according to the sizes of a and μ: either both are ≥ 12 (Case E), or $\mu \leq 11$ (Case

T), or $\mu \geq 12$ and $a \leq 11$ (Case B). Each branch delivers a multiset of size $\leq M-1$ whose subset sums contain M consecutive integers. □

We can now also discharge the sharpness claims stated in the introduction.

Proof of Proposition 1.5. The multiset $\{3,3,4,5\}$, of size 4, has subset sums $\{0,3,4,5,6,7,8,9,10,11,12,15\} \supseteq [3,12]$, a run of $10 \geq 5$ consecutive integers, so $m \leq 4$. For the lower bound one checks the finitely many multisets drawn from $\{3,4,5\}$ of size at most 3. There are 19 nonempty such multisets (3 of size 1, 6 of size 2, 10 of size 3), and none has 5 consecutive integers among its subset sums; the empty multiset, with subset sums $\{0\}$, is trivially excluded. In fact the longest subset-sum run achievable at size at most 3 is only 3, attained by $\{3,4,5\}$ itself, whose subset sums are $\{0,3,4,5,7,8,9,12\}$ (longest run $[3,5]$). Hence $m(\{3,4,5\}) = 4 = M-1$. For the cardinality claim, the five multisets drawn from $\{2,3\}$ of size ≤ 2 have subset-sum sets $\{0,2\}$, $\{0,3\}$, $\{0,2,4\}$, $\{0,2,3,5\}$, $\{0,3,6\}$, none with a run of 3. □

5. Assembly: proof of Theorem A

Proof of Theorem A. Part (1) is Theorem 2.1.

For part (2), let $d_2 \leq k$ with $k \geq 3$, and let (r_i) be any prescribed growth sequence. Build the certificate B of Lemma 3.1. Let A be any (d_1,d_2)-sequence, let G_∞ be its tail alphabet and $g := \gcd(G_\infty)$. By Lemma 3.3 we may pass to the tail, and by Lemma 3.5 we may assume $\gcd(G_\infty) = 1$ after rescaling. Then, retracing the map of §1.7:

Case	Handled by
$\lvert G_\infty/g\rvert = 1$	Lemma 3.4 (AP tail)
$\lvert G_\infty/g\rvert = 2$, gap word eventually periodic	Lemma 3.6
$\lvert G_\infty/g\rvert = 2$, k even, $d_2 = k$ (boundary)	Proposition 3.10
$\lvert G_\infty/g\rvert = 2$, some $W_2(\sigma_0) \geq 2$, k odd or $d_2 \leq k-1$	Lemmas 3.9(a) and 3.8
$\lvert G_\infty/g\rvert = 2$, all $W_2 \leq 1$ (balanced)	Lemma 3.9(b): Prop. 3.11, then Lemma 3.6 or 3.14
$\lvert G_\infty/g\rvert \geq 3$	Lemma 3.18 (slot lemma), with $m(G_\infty/g) \leq \max(G_\infty/g) - 1 \leq d_2 - 1 \leq k-1$ by Lemma 1.4

In every case kA is tail-covering (modulo $g \cdot m'$, after Lemma 3.5) — this is Proposition 3.20 — so B meets kA by Lemma 3.1. Since the growth sequence (r_i) was arbitrary, non-existence holds in the strong form asserted.

Combining (1) and (2): if $d_2 \geq k+1$ then $r_k(d_1,d_2)$ exists, and if $d_2 \leq k$ it does not. As exactly one of these holds for any k, d_2, the dichotomy follows. □

6. Formal verification in Lean 4

The entire dichotomy has been formalized in Lean 4 against Mathlib, and machine-checked. The development, the verification scripts and build instructions are available at `https://github.com/beetree/math_erdos_1112`. This section records the formal statement, argues its faithfulness to the informal problem, and states the trust base. Appendix C maps each load-bearing result above to its Lean declaration.

Because a formal proof certifies only its stated theorem, the encoding is given in full and justified below. The definitions live in a frozen statement file `Erdos1112.lean`, which is imported by the development but never edited by it; the three theorems are stated once each, directly over those definitions, in `Erdos1112Proof/Final.lean`. There is a single copy of each definition, so what is audited below is literally what the theorems assert.

6.1. The encoding. Sequences A, B are functions $\mathbb{N} \to \mathbb{N}$, 0-indexed (so `b 0` is b_1).

```
def IsLacunaryWith (r : ℕ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, r * b i ≤ b (i + 1)

def HasGapsIn (d₁ d₂ : ℕ) (a : ℕ → ℕ) : Prop :=
  0 < a 0 ∧ ∀ i, a i + d₁ ≤ a (i + 1) ∧ a (i + 1) ≤ a i + d₂

def kFoldSumset (k : ℕ) (a : ℕ → ℕ) : Set ℕ :=
  { n | ∃ f : Fin k → ℕ, n = ∑ j, a (f j) }

def RatioWorks (k d₁ d₂ r : ℕ) : Prop :=
  ∀ b : ℕ → ℕ, IsLacunaryWith r b →
    ∃ a : ℕ → ℕ, HasGapsIn d₁ d₂ a ∧
      Disjoint (kFoldSumset k a) (Set.range b)

def Question (k d₁ d₂ : ℕ) : Prop := ∃ r : ℕ, RatioWorks k d₁ d₂ r

def IsVarLacunaryWith (R : ℕ → ℕ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, R i * b i ≤ b (i + 1)
```

These are reproduced verbatim from the source. The three theorems are then, in `Final.lean`:

```
theorem erdos_1112 (k d₁ d₂ : ℕ) (hk : 3 ≤ k) (hd₁ : 1 ≤ d₁) (hd : d₁ < d₂) :
    Question k d₁ d₂ ↔ k + 1 ≤ d₂

theorem erdos_1112_existence_bound (k d₁ d₂ : ℕ) (hk : 3 ≤ k) (hd₁ : 1 ≤ d₁)
    (hd : d₁ < d₂) (h : k + 1 ≤ d₂) : RatioWorks k d₁ d₂ (192 * d₂)

theorem erdos_1112_strong_nonexistence (k d₁ d₂ : ℕ) (hk : 3 ≤ k)
    (hd₁ : 1 ≤ d₁) (hd : d₁ < d₂) (h : d₂ ≤ k) (R : ℕ → ℕ) :
    ∃ b : ℕ → ℕ, IsVarLacunaryWith R b ∧
      ∀ a : ℕ → ℕ, HasGapsIn d₁ d₂ a →
        (kFoldSumset k a ∩ Set.range b).Nonempty
```

6.2. Why the encoding is faithful. A handful of modeling choices carry the weight.

(1) **Infinite increasing sets as strictly increasing functions.** A and B are infinite sets of positive integers, encoded by their increasing enumerations. `StrictMono` is what makes the function enumerate a *set* rather than a multiset; it is imposed on B directly and *derived* for A from the gap condition when $1 \leq d_1$ (the lemma `HasGapsIn.strictMono`).

(2) **The gap window, phrased additively.** Over $\mathbb{N}$, subtraction is truncated. The definition writes $d_1 \leq a_{i+1} - a_i \leq d_2$ as $a_i + d_1 \leq a_{i+1} \wedge a_{i+1} \leq a_i + d_2$. This avoids a real trap: with a subtractive phrasing, on a *decreasing* step $a_{i+1} - a_i$ underflows to 0, so an isolated upper bound $a_{i+1} - a_i \leq d_2$ would read $0 \leq d_2$ — true — silently admitting negative gaps. (The lower bound is not the trap: an underflowed gap gives $d_1 \leq 0$, false, since $1 \leq d_1$.)

(3) **The sumset, with repetition.** `kFoldSumset` ranges over an arbitrary index function $f : \mathrm{Fin}\, k \to \mathbb{N}$ with *no* injectivity requirement, so summands may repeat. Requiring f injective would model the distinct-summand sumset, a strictly smaller set, and a different problem.

(4) **Quantifier order.** `Question` is $\exists r\ \forall B\ \exists A$, with A chosen after B — not the easier $\exists A\ \forall B$.

(5) **The ratio over $\mathbb{N}$ rather than $\mathbb{Z}$.** The problem says "an integer r". Passing to $\mathbb{N}$ loses no existence content: a natural witness is an integer witness; and given an integer

witness $r < 0$, the ratio condition is vacuous (as B is positive, $rb_i \le 0 < b_{i+1}$), so the admissible class of B coincides with that at $r = 0$, and 0 is a natural witness. The lemma `RatioWorks.mono` — a larger r only shrinks the admissible class of B — is one machine-checked ingredient. The full $\mathbb{Z} \to \mathbb{N}$ bridge is now also formalized: the frozen file carries the integer form `QuestionInt` and the theorem `question_iff_questionInt`, that `Question` $\leftrightarrow$ `QuestionInt`, so the reduction is a proved equivalence, not only an informal justification (§6.4).

(6) **A genuine, non-vacuous biconditional.** Both directions are separately witnessed, and both sides of the iff are inhabited within the allowed parameter family: when $d_1 \le k$, the boundary $d_2 = k + 1$ realizes the existence side and any $d_1 < d_2 \le k$ realizes the non-existence side. None of the predicates is silently trivial: `IsLacunaryWith r` is satisfiable for every r (take $b_i = (r+2)^i$), `HasGapsIn` is satisfiable (take $a_i = 1 + d_1 i$), and `kFoldSumset k a` is nonempty while `Set.range b` is infinite, so the disjointness requirement is a genuine constraint.

(7) **Strong non-existence.** `erdos_1112_strong_nonexistence` says strictly more than $\neg$`Question`: for *every* ratio sequence R there is a *single* B, lacunary with those varying ratios, colliding with the sumset of every admissible A. Specializing $R \equiv r$ recovers $\neg$`RatioWorks` $k\ d_1\ d_2\ r$ for every r, hence $\neg$`Question`; the bridge is the lemma `isVarLacunaryWith_const_iff`.

6.3. **Trust base.** We state the claim in the form in which it is checkable. *At the repository commit recorded in §9, built with the pinned toolchain `leanprover/lean4:v4.27.0` and the Mathlib revision fixed by `lake-manifest.json`, the command `#print axioms` reports, for each of the three headline theorems, exactly the list below.* The development is `sorry`-free: a source scan finds no `sorry`, no `axiom` declaration, and no `native_decide` anywhere. Accordingly the audit reports

```
[propext, Classical.choice, Quot.sound]
```

— the three standard foundational axioms of Lean and Mathlib, and nothing else. In particular there is no `sorryAx` (no gaps) and no `Lean.ofReduceBool`, the latter being the axiom that `native_decide` would introduce. Every finite and decidable check — the certificate tables of Proposition A.1 and the $a \le 3000$ Case-T scan — is therefore closed by the trusted *kernel*, not by compiled code: the trusted base is Lean plus Mathlib, and not the Lean compiler. The toolchain is pinned (`leanprover/lean4:v4.27.0`, with Mathlib pinned via `lake-manifest.json`), so the audit is reproducible from a clean checkout:

```
lake build                                        # compiles the development
lake env lean Erdos1112Proof/AxiomsCheck.lean # the 3-axiom list, per theorem
```

The axiom audit is itself the rigorous gap check: any `sorry` would surface as `sorryAx` in the reported list, and any `native_decide` as `Lean.ofReduceBool`; neither appears.

Two heavyweight declarations set a raised kernel heartbeat locally (`tailCovering_of_sturmian` and `exists_safe_subinterval`); this is an elaboration budget only, and carries no trust cost.

A caveat on what this does and does not say. "Depends only on three axioms" is a statement about the *logical* axiom report. The operative trusted base is larger: it includes the Lean kernel and elaborator, the Mathlib source at the pinned revision, and the dependency snapshot. And an axiom audit cannot detect a mismatch between a numbered lemma in this paper and the declaration that purports to formalize it; that correspondence is asserted in Appendix C and must be checked by reading both. Where the formal route differs from the prose route we say so explicitly (Remark 4.16 and §6.5).

6.4. **Scope of the formal development.** We record two points about scope, because the reader is entitled to know exactly what the machine checks — and, in one case, what it deliberately does not.

(1) **The $\mathbb{Z} \to \mathbb{N}$ bridge.** The problem asks for an *integer* r; the development quantifies r over $\mathbb{N}$. The frozen statement file now also carries the integer form,

```
def IsLacunaryWithInt (r : ℤ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, r * (b i : ℤ) ≤ (b (i + 1) : ℤ)

def QuestionInt (k d₁ d₂ : ℕ) : Prop := ∃ r : ℤ, RatioWorksInt k d₁ d₂ r
```

together with the machine-checked equivalence `question_iff_questionInt`, that `Question` ↔ `QuestionInt`, and the headline theorem in integer form,

$$\texttt{erdos\_1112\_int}: \quad \texttt{QuestionInt}\ k\ d_1\ d_2\ \leftrightarrow\ k+1 \leq d_2.$$

So the dichotomy is verified for the problem's literal integer phrasing, not merely for a natural-number rendering of it.

(2) **The case $k = 2$ of Theorem 2.1.** Theorem 2.1 is proved here in prose for all $k \geq 2$, but the formalization covers $k \geq 3$ only — `existence_bound` and `erdos_1112_existence_bound` carry the hypothesis $3 \leq k$. This is deliberate rather than a gap to be closed: the problem fixes $k \geq 3$, so the formal development carries exactly the problem's hypotheses throughout, and the $k = 2$ existence statement stands as an unformalized supplementary observation (Remark 2.3).

The verification claim is therefore: *Theorem A for $k \geq 3$, in both the natural and the integer phrasing, is machine-checked over the definitions of §6 at the release named in §9.* Every headline Lean theorem carries the problem's own hypothesis $3 \leq k$. Nothing in the mathematical argument of this paper is delegated to the formalization.

6.5. **Deviations of the formalization from this text.** The current paper and formalization agree on every mathematical statement. The deviations below are differences of *route* — the formalization sometimes proves the same conclusion a different way — not disagreements about what is true. (During development, formalization did expose a missing hypothesis in an earlier draft of Lemma 4.3(d); the present statement includes it, and its necessity is witnessed by the counterexample in Remark 4.4.) They have all been incorporated into the text above.

(1) Case P is proved by one r-parameterized construction (Lemma 4.14), replacing separate $r = 3$ and $r = 4$ families, three ad-hoc $r = 5$ certificates, and a finite check for $a < 15$. The reach inequalities (19) close algebraically for every $a \geq 3$ and $r \geq 3$ satisfying (20), and the only triple escaping that estimate is $(3, 4, 5)$, checked inline. No subset-sum search occurs in Case P, in the text or in the formalization, and no certificate table is needed. In Lean the construction is `caseP_pair` (uniform in r, taking (19) as hypotheses), instantiated by `caseP_r3` / `caseP_r4` and completed by `caseP_large`; see Remark 4.16.

The route uses the following independent argument for the range $M \geq 2a + 4$, which is shorter there; in the text, Lemma 4.14 alone closes the whole case.

Lemma 6.1. *If $a \mid b + M$ and $M \geq 2a + 4$, then the hard-core triple (a, b, M) admits a multiset of budget $\leq M - 1$ whose subset sums contain M consecutive integers.*

Proof. Since $\bar{\mu} = a - \bar{e}$, the representatives $jb + kM \equiv \bar{e}(j-k) \pmod{a}$ cover $\mathbb{Z}_a$ with any box (Y, Z) satisfying $Y + Z = a - 1$, as $j - k$ then ranges over $[-Z, Y]$, a complete residue system. Choose $Y = \lceil (a-1)/2 \rceil$ and $Z = \lfloor (a-1)/2 \rfloor$. The minimal representative of the class $\bar{e}w$ has height wb (for $w \geq 0$) or $|w|M$ (for $w < 0$), so $S \leq \max(Yb,\ ZM)$, and the budget satisfies $\lceil (M - 1 + S)/a \rceil + (a - 1) \leq M - 1$ iff $S \leq (a-1)M - a^2 + 1$.

The ZM branch gives $ZM \le \frac{a-1}{2}M \le (a-1)M - a^2 + 1$ once $M \ge 2a+2$. For the Yb branch the parity of a enters. If a is odd then $Y = Z = (a-1)/2$ and the same computation applies. If a is even then $Y = a/2$, and the bound $b \le M-1$ alone would require $M \ge 2a + 3 + \frac{4}{a-2}$, which exceeds $2a+4$ at $a = 4$. But $b \le M-1$ is not tight on this line: $a \mid b + M$ forces $b + M \equiv 0 \pmod{a}$, so for even a the sum $b+M$ is even, whence $b \ne M-1$ and therefore $b \le M-2$. With $b \le M-2$ the Yb branch closes for $M \ge 2a+4$ as well. (In particular the a-even, $b = M-1$ corner is vacuous, not merely bounded.) □

(2) Case D closes uniformly on $(a-1)(q-2) \ge 0$, with no split on q (Lemma 4.13).
(3) The two-letter core takes a density-free route: the balanced classification via a discrepancy-oscillation iteration (Proposition 3.11), and the uniform hitting of Lemma 3.13 via an explicit Dirichlet-step circle walk, rather than the three-distance theorem. Dirichlet's approximation theorem is the only analytic input.
(4) The free-gap lemma is proved at the single index $s_1 = \lceil t/\mathrm{hi} \rceil$ (where hi is the upper endpoint of the parent interval; cf. Lemma 2.2), with safety for all s by monotonicity (Lemma 2.2).
(5) Case T is run through the two merge-robust staircase variants only, in the text and in the formalization alike; neither uses Lemma 4.3(c). Table A is the failure set of that route, and is the only certificate table Case T needs.
(6) The strong non-existence theorem is stated in the constructive `Set.Nonempty` form.

Formalizing the tables and scans exposed no erroneous, missing or redundant object. The extended form of Lemma 4.3(d), whose pre-repair statement was false, is carried in its repaired form, and its counterexample $(5, 7, 9)$ (Remark 4.4) is pinned as a decided guard.

7. Provenance and authorship

This work was produced with substantial assistance from automated systems, and we record the division of labour.

The mathematical constructions, case analyses, and proofs, together with the Lean formalization and a first draft of this paper, were produced by Claude (Anthropic; models Fable 5 and Opus 4.8) acting as the reasoning agent, under the author's direction. GPT-5.5 (OpenAI) and Gemini 3.1 (Google DeepMind) were used for advice and for adversarial review of drafts.

The author selected the problem and fixed its formalization (the Erdős Problems phrasing of §1); set the intermediate targets and overall strategy; decided which candidate routes to pursue and which to abandon; audited each construction and proof; chose the formal statement; and made the final mathematical judgments. He takes full responsibility for the content, including any error that remains, and can defend each part of the argument. The automated systems are tools, not authors: they meet no authorship criterion, cannot take responsibility for the work, and are not listed as authors.

Formal verification settles correctness relative to the formal statement only; it does not establish novelty, attribution, or exposition. The development is machine-checked in Lean 4 (§6), and the finite certificates can be re-verified independently (Appendix A). The one thing no formalization can settle on the reader's behalf — that the formal statement means what the informal problem says — is set out decision by decision in §6.

8. Declarations

Funding. None.
Competing interests. The author declares no competing interests.

Author contributions. Sole author. The division of labour between the author and the automated systems used is set out in §7.

Use of generative AI. Automated systems played a substantial role in producing this work; the full account — models, tasks, division of labour, and verification — is §7. The author directed the work, audited the output, and takes full responsibility for the content; the systems are tools, not authors. Every mathematical claim used in the proof of Theorem A has been checked by at least one of the three routes of §1.7, and the dichotomy is machine-checked in Lean 4 within the scope of §6.4.

A note on repository metadata. The version-control history of the accompanying repository contains `Co-Authored-By` trailers naming the automated system. These record which tooling produced a commit; they are not a claim of scholarly authorship, which is as stated above.

Data and code availability. See §9.

9. Data availability and reproduction

Everything needed to check this paper is contained in one commit of the repository `https://github.com/beetree/math_erdos_1112` — the commit to which the annotated tag `submission-2026-07` resolves. That single commit carries this LaTeX source (article and supplement), the Lean 4 development [Lean4, Mathlib], the machine-readable certificate exports (`table-A.csv`, `table-B.csv`, `SHA256SUMS`), the table generator/re-verifier `gen-tables.py`, and the two Python harnesses. All checks below refer to that commit.

The machine-checked proof. A clean checkout builds with the pinned toolchain (`leanprover/lean4:v4.27.0`; dependency revisions fixed in `lake-manifest.json`): `lake exe cache get` then `lake build` compiles every file and reproduces the axiom audit of §6.3, including for `erdos_1112_int`, the literal integer form. Continuous integration at the repository (`.github/workflows/verify.yml`) re-runs the pinned build, the axiom audit, the table re-verification, and the correspondence check of Appendix C on every push; the public run logs serve as clean-build evidence.

The finite layer. The complete certificate tables are typeset in the supplementary document `erdos1112-supplement.pdf` and exported as `data/table-A.csv` and `data/table-B.csv` (digests in `data/SHA256SUMS`); the article itself gives the definition and a worked example of each (appendix on the certificate tables). Each row is a self-certifying subset-sum witness, mechanically verifiable from its tuple, and the Lean kernel re-decides all of them in the build above. Running `make` in `paper/` regenerates both the supplement and the exports from the one canonical source and re-checks every row (budget $\leq M-1$ and coverage of M consecutive integers, by exact bitmask, aborting on any failure), so no listing — printed, exported, or kernel-decided — can silently diverge. The two Python harnesses of Appendix A reproduce the branch counts and scans of Proposition A.1.

Submission artifact. This version is the annotated Git tag `submission-2026-07` of the repository `https://github.com/beetree/math_erdos_1112`. The article and supplement sources, the certificate data, the generator/re-verifier, the Python harnesses, and the Lean 4 development are the *content commit*

`@CONTENT-SHA@`

to which the tagged commit is the immediate successor, byte-identical except for embedding this identifier and the rebuilt PDFs (a commit cannot contain its own hash). The axiom audit of §6.3 and the continuous-integration checks run at both commits. This PDF is built from the tagged commit by a clean checkout; a DOI-backed archive of the same release will accompany the journal submission for scholarly citation. The full certificate tables are typeset in the supplement `erdos1112-supplement.pdf` and exported to `data/`.

Appendix A. The finite layer

We isolate the entire finite content of the proof as a single statement, so that it is auditable at once rather than scattered through the argument.

Proposition A.1 (Finite verification)**.** *The finite certificate set — Table A, Table B and the Table-B supplement — has the following properties, each a finite, mechanical check:*

(1) *every certificate has budget* $\le M-1$*;*
(2) *the subset sums of every certificate contain* M *consecutive integers;*
(3) *the Case-T line scan (*$\mu \le 11$*) over* $a \le 3000$ *produces exactly the* 172 *exceptional triples of Table A, and none for* $a > 3000$*;*
(4) *the Case-B base classes (*$a \le 11$*) are complete;*
(5) *the* λ*-lift (Lemma 4.5) applies to every larger member of each Table-B class.*

Proof. (i)–(ii) are direct finite checks: for each row, sum the budget and compute its subset-sum set by exact arbitrary-precision bitmask, verifying that it contains a run of length M. This is where Lemma 4.2 enters: a modulo-a box with $Y + Z \le a - 1$ certifies exactly such a run. (iii)–(iv) are exhaustive bounded scans — the $a \le 3000$ line scan of Proposition 4.22 and the $a \le 11$, bounded-b class enumeration of §4.9 — made complete by the slope-tail argument in the proof of Proposition 4.22 for $a > 3000$ and by the descent of Lemma 4.24 for larger b. Finally (v) is Lemma 4.5. Note that Case P contributes nothing to this proposition: by Remark 4.16 it is closed in the text, with no certificate. □

A.1. **What is checked, and how.** Three kinds of computation participate, and no others.

Finite certificate tables (Appendix B): $172+178$ explicit instances of the proved frame lemma (Lemma 4.2). Each is checked by exact subset-sum computation with arbitrary-precision bitmasks, and Lemma 4.5 extends each Table-B certificate to the infinitely many larger members of its class. Case P contributes *no* certificates (Remark 4.16).

Finite scans closing crude constants of proved lemmas: the Case-T line scan to $a = 3000$ with the slope-tail argument (Proposition 4.22); and the minimal-alphabet enumeration to $M \le 130$ (Lemmas 4.11 and 4.12).

End-to-end redundant verification (not logically required, but performed): every one of the 83,251 hard-core triples with $M \le 120$ was covered by its designated branch with an exactly verified multiset — zero failures — by two separate implementations. The first, `sharp6_final.py`, is the constructive harness. The second, `sharp_referee_check.py`, is a re-implementation that shares no code with the first and reconstructs each designated witness from the lemma statements alone, with no auxiliary searches; it reports identical branch counts, and additionally re-verifies Lemmas 4.1–4.5 and 4.7–4.10 by brute force on thousands of instances, re-validates both tables including base-minimality and class-completeness, walks the λ-lift chains over 29,402 members arithmetically to $M \le 1500$ and 4,574 exactly to $M \le 250$, checks the pair families for all $a \le 120$, and checks the η-box algebra to $a \le 400$ at the worst member $M = a + 12$ (which, being monotone in M, confirms Lemma 4.18 for all M). All with zero failures. A further 800 random hard-core triples with $M \in (120, 200]$ were spot-checked against the designated branches, again with zero failures.

Scope of the independence. We should not overstate what the second harness buys. It is best described as an *independent implementation of the same certificate specification*: it shares no code with the first and rebuilds each witness from the lemma statements, but it is *not* independent of the mathematical case split, the witness design, or the branch structure — all of which are the ones stated in §4. Agreement between the two therefore rules out coding slips, not errors in the case analysis itself. It corroborates the finite layer only, and says nothing about the asymptotic lemmas, which are established in the text and formalized in Lean.

A.2. **Kernel re-checking of the finite layer.** The formalization re-verifies the entire finite layer inside the Lean kernel, with no `native_decide` and no external process, so the Python harnesses are corroborated rather than relied upon.

The printed tables and the kernel-decided data now agree exactly: 172 Case-T rows and 178 distinct Case-B classes, 350 certificates in all, with Case P contributing none (it is closed in prose). The prose proof and the formal proof use the same lemmas, the same merge-robust Case-T variants (Lemma 4.20), and the same tables; there is no route divergence in the certificate data. The only bookkeeping detail is internal to the Lean sources: the 172 Table-A rows are stored as two lists — `certTableA` (158 rows) and a 14-row supplement `tSuppT` that discharges the merge-robust variant — whose union is the printed Table A. All of this data is exported in machine-readable form (§9).

The $a \leq 3000$ Case-T scan is likewise redone as kernel `decide` sweeps, in chunks of 1000 values.

Finally, the certificate tables (typeset in full in the supplement, exported to `data/`, and exemplified in Appendix B) are generated mechanically from the same source data that the Lean development transcribes, and every row is re-verified at generation time (budget and coverage, by exact bitmask), so no listing can silently diverge from the checked one.

Appendix B. The certificate tables

Each entry is a hard-core triple (a, b, M) together with a *box certificate* (x, Y, Z), asserting that the multiset S consisting of x copies of a, Y copies of b and Z copies of M has

- **budget** $x + Y + Z \leq M - 1$, and
- **coverage:** subset-sums(S) contains M consecutive integers,

i.e. that S witnesses $m(G) \leq M - 1$ for $G = \{a, b, M\}$. The mechanism is Lemma 4.2: the copies of b and M realize a modulo-a box (Y, Z) with $Y + Z \leq a - 1$ covering $\mathbb{Z}_a$, and the x copies of a lift that residue cover to a solid run $[S,\ S + M - 1]$.

Example B.1. Row $(5, 8, 9) \to (5, 2, 1)$: the multiset $\{5, 5, 5, 5, 5,\ 8, 8,\ 9\}$ has budget $5 + 2 + 1 = 8 = M - 1$. Its subset sums $\{5i + 8j + 9k : i \leq 5,\ j \leq 2,\ k \leq 1\}$ hit every residue modulo 5 — the parts $8j + 9k \in \{0, 8, 16, 9, 17\}$ have residues $\{0, 3, 1, 4, 2\}$ — and, lifted by the copies of 5, contain the run $[17, 25]$, which is $9 = M$ consecutive integers. That single computation is exactly what the checker performs on each row.

Table A lists the 172 exceptional lines of Case T (Proposition 4.22), all with $a \leq 29$. Table B lists one base per class $(a, \bar{e}, h)$ for Case B, extended to all larger members by Lemma 4.5 (the Lean transcription additionally re-bases the six diagonal classes of Lemma 4.24). A representative Table-B row is the class $[5; 1, 1]$ with base $(a, b, M) = (5, 16, 17)$ and box $(x, Y, Z) = (10, 1, 2)$: budget $10 + 1 + 2 = 13 \leq M - 1 = 16$, and the parts $16j + 17k$ ($j \leq 1$, $k \leq 2$) realize every residue modulo 5, which the copies of 5 lift to a run of $17 = M$ consecutive integers (cf. Example 4.23).

Rather than fill several pages of the article with $172 + 178$ dense rows — each an independently machine-checkable tuple, of little value read by eye — we provide the *complete* tables as supplementary material: the typeset listing `erdos1112-supplement.pdf` (built by the same `make`), and the machine-readable exports `data/table-A.csv` and `data/table-B.csv`, whose SHA-256 digests are recorded in `data/SHA256SUMS`. All are generated from the one canonical source and re-verified row by row by `gen-tables.py` (§9), and the same rows are decided by the Lean kernel (§6.3); the tables are thus present in full, simply not inlined here.

Appendix C. Correspondence with the Lean development

Declarations live in the namespace `Erdos1112.Proof`, in files relative to `Erdos1112Proof/`, except the three headline theorems, which are in namespace `Erdos1112`. Every listed result is load-bearing: the development formalizes the full case tree rather than citing anything externally.

The name-and-location layer of this table is checked mechanically: the script `scripts/check_correspondence.` extracts every declaration named below and verifies it exists in the Lean tree, and continuous integration re-runs the check on every push, so the table cannot silently rot as the development evolves. The *statement-level* match — that each declaration says what the prose result says — remains a human judgment, as §6.3 already cautions, and is where we invite scrutiny.

The mapping is *not* a claim of lemma-by-lemma identity everywhere. In one place the formal development proves the same conclusion by a different route, and we mark those rows ALT: Case P, where the paper gives a single uniform argument for the whole line (Lemma 4.14) while Lean instantiates $r \in \{3, 4\}$ and routes the rest through Lemma 6.1 (Remark 4.16). The formal statement proved is the one the assembly consumes; it is not a weaker result. Everywhere else — including Case T, which now uses the same merge-robust route in the text and in Lean — the prose and the formalization follow the same argument.

Result here	Lean declaration	File
Theorem A (dichotomy)	`erdos_1112`	`Final.lean`
Theorem 2.1	`erdos_1112_existence_bound`, `existence_bound`	`Final.lean`, `Existence/Nested.lean`
Strong non-existence	`erdos_1112_strong_nonexistence`	`Final.lean`
Theorem A, integer form ($\mathbb{Z} \to \mathbb{N}$ bridge)	`erdos_1112_int`, `question_iff_questionInt`	`Final.lean`, `Erdos1112.lean`
Lemma 2.2 (free gap)	`exists_safe_subinterval`	`Existence/FreeGap.lean`
Lemma 3.1 (certificate)	`cert_hit`	`NonEx/Certificate.lean`
Lemmas 3.4, 3.6	`tailCoveringN_of_single_letter`, `..._of_eventually_periodic`	`NonEx/GapWord.lean`
Lemma 3.7 (interval)	`Wset_interval`	`NonEx/TwoLetter/Core.lean`
Lemma 3.8 (sweep)	`sweep`	`NonEx/TwoLetter/Core.lean`
Lemma 3.9 (width dichotomy)	`width_of_unbalanced`	`NonEx/TwoLetter/Core.lean`

Prop. 3.11 (balanced)	`balanced_classification`
	`disc_osc_le_one`, `mechanical_tail`
	`slope`, `slope_le_one`
		`NonEx/TwoLetter/Balanced.lean`
		`NonEx/TwoLetter/MH/IrrationalCase.lean`
		`NonEx/TwoLetter/MH/Slope.lean`

Lemma 3.12 (boundary)	`width_even_boundary`	`NonEx/TwoLetter/Core.lean`
Lemma 3.14 (Sturmian)	`tailCovering_of_sturmian`, `walk_enters`	`NonEx/TwoLetter/Sturmian.lean` `NonEx/TwoLetter/MH/Walk.lean`
Lemma 3.18 (slot)	`tailCovering_of_three_letters`	`NonEx/SlotLemma.lean`
Lemma 4.1 (two-generator)	`twoGen_interval`, `twoGen_hasRun`	`Sharp/TwoGen.lean`
Lemma 4.2 (frame)	`frame_lemma`	`Sharp/Frame.lean`
Lemma 4.3 (staircase)	`staircase_phase_base`, `staircase_phase_extended`	`Sharp/Staircase.lean`
Lemma 4.5 (λ-lift)	`FrameCert.lift`	`Sharp/Lift.lean`
Lemma 4.6 (Graham)	`sharpAt_of_hardcore`	`Sharp/Graham.lean`
Lemma 4.12 (minimal alphabets)	`sharp_of_minimal`	`Sharp/Graham.lean`
Lemma 4.13 (Case D)	`caseD`	`Sharp/CaseD.lean`
Lemmas 4.14, 6.1 (Case P) ALT (see above)	`caseP_pair`, `caseP_r3`, `caseP_r4`, `caseP_large`	`Sharp/CaseP.lean`
Lemma 4.17 (Case L)	`caseL`	`Sharp/CaseL.lean`
Lemma 4.18 (Case E)	`caseE`	`Sharp/CaseE.lean`
Lemma 4.20, Prop. 4.22 (Case T)	`caseT`, `T_tail_line`, `tSuppT`	`Sharp/CaseT*.lean`
Case B (+ descent)	`caseB`, `rowFor`, `caseBComplete`	`Sharp/CaseB.lean`, `Sharp/CaseBClasses.lean`
Tables A/B (data)	`certTableA_ok`, `certTableB_ok`	`Sharp/TablesData.lean`
Lemma 1.4 (bounded subset-sum)	`sharp`	`Sharp/Main.lean`
Axiom audit	three `#print axioms` commands	`AxiomsCheck.lean`

Index of coined terms

For linear readers and auditors alike: each term of art used in this paper, with its definition site.

Term	Defined	Term	Defined
(d_1, d_2)-sequence	§1	frame	Lemma 4.2
tail-covering	§1.8	budget $\mathcal{B}$	after Lemma 4.2
gap word, tail alphabet G_∞	§1.8	level, staircase	Lemma 4.3
balanced, Sturmian, mechanical	§3.3	hard core	§4.3
antidiagonal width W_2	§3.3	line (Case T)	§4.8
congruence-compatible	Lemma 3.14	class, class base	§4.9
fine slot, coarse dial	Lemma 3.18	λ-lift, descent	Lemmas 4.5, 4.24
mover (Case P)	§4.5	merge-robust	Lemma 4.20
Construction T	§4.8	box certificate	Appendix B

References

[AF88] N. Alon and G. Freiman, *On sums of subsets of a set of integers*, Combinatorica **8** (1988), no. 4, 297–306.

[BHJ97] B. Bollobás, N. Hegyvári and G. Jin, *On a problem of Erdős and Graham*, Discrete Math. **175** (1997), no. 1–3, 253–257. DOI: 10.1016/S0012-365X(96)00122-7.

[CFP21] D. Conlon, J. Fox and H. T. Pham, *Subset sums, completeness and colorings*, preprint, arXiv:2104.14766 (2021).

[CMZ25] L. Chen, Y. Mao and G. Zhang, *Long arithmetic progressions in sumsets and subset sums: constructive proofs and efficient witnesses*, in: Proc. 57th Annual ACM Symposium on Theory of Computing (STOC 2025), ACM, 2025, 2086–2097. DOI: 10.1145/3717823.3718281.

[Ch00] Y.-G. Chen, *On sums and intersects of sequences*, Discrete Math. **223** (2000), no. 1–3, 351–354. DOI: 10.1016/S0012-365X(00)00097-2.

[E1112] T. F. Bloom, *Erdős problem #1112*, `https://www.erdosproblems.com/1112`. Accessed 13 July 2026.

[ErGr80] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique **28**, Université de Genève, Geneva, 1980, 128 pp.

[Gr64] R. L. Graham, *Complete sequences of polynomial values*, Duke Math. J. **31** (1964), no. 2, 275–285. DOI: 10.1215/S0012-7094-64-03126-6.

[GS20] A. Granville and G. Shakan, *The Frobenius postage stamp problem, and beyond*, Acta Math. Hungar. **161** (2020), no. 2, 700–718. DOI: 10.1007/s10474-020-01073-y.

[GW21] A. Granville and A. Walker, *A tight structure theorem for sumsets*, Proc. Amer. Math. Soc. **149** (2021), no. 10, 4073–4082. DOI: 10.1090/proc/15608.

[Lean4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction — CADE 28, Lecture Notes in Computer Science **12699**, Springer, 2021, 625–635.

[Lev03] V. F. Lev, *Blocks and progressions in subset sum sets*, Acta Arith. **106** (2003), no. 2, 123–142.

[Lev22] V. F. Lev, *The structure of higher sumsets*, Proc. Amer. Math. Soc. **150** (2022), no. 12, 5165–5177. DOI: 10.1090/proc/16128. (arXiv:2110.03554.)

[Lev98] V. F. Lev, *On consecutive subset sums*, Discrete Math. **187** (1998), no. 1–3, 151–160.

[Lo02] M. Lothaire, *Algebraic Combinatorics on Words*, Encyclopedia of Mathematics and its Applications **90**, Cambridge University Press, 2002. (Ch. 2: Sturmian words, balancedness and the Morse–Hedlund theorem.)

[Mathlib] The mathlib Community, *The Lean mathematical library*, in: Proc. 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 367–381.

[MH40] M. Morse and G. A. Hedlund, *Symbolic dynamics II. Sturmian trajectories*, Amer. J. Math. **62** (1940), no. 1, 1–42. DOI: 10.2307/2371431.

[Na72] M. B. Nathanson, *Sums of finite sets of integers*, Amer. Math. Monthly **79** (1972), no. 9, 1010–1012. DOI: 10.1080/00029890.1972.11993172.

[RA05] J. L. Ramírez Alfonsín, *The Diophantine Frobenius Problem*, Oxford Lecture Ser. Math. Appl. **30**, Oxford Univ. Press, 2005.

[Sa89] A. Sárközy, *Finite addition theorems, I*, J. Number Theory **32** (1989), no. 1, 114–130.

[Sa94] A. Sárközy, *Finite addition theorems, II*, J. Number Theory **48** (1994), no. 2, 197–218.

[SV06] E. Szemerédi and V. H. Vu, *Finite and infinite arithmetic progressions in sumsets*, Ann. of Math. (2) **163** (2006), no. 1, 1–35. DOI: 10.4007/annals.2006.163.1.

[TaYa21] M. Tang and Q.-H. Yang, *On a problem of Erdős and Graham*, Publ. Math. Debrecen **99** (2021), no. 3–4, 485–493. DOI: 10.5486/PMD.2021.9045.

Independent researcher
Email address: johan.land@gmail.com